arXiv:2512.19155v1 [cs.AI] 22 Dec 2025

Can We Test Consciousness Theories on AI? Ablations, Markers, and Robustness

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December 23, 2025

Abstract

The search for reliable indicators of consciousness has fragmented into competing theoretical camps (Global Workspace Theory (GWT), Integrated Information Theory (IIT), and Higher-Order Theories (HOT)), each proposing distinct neural signatures. We adopt a synthetic neuro-phenomenology approach: constructing artificial agents that embody these mechanisms to test their functional consequences through precise architectural ablations impossible in biological systems. Across three experiments, we report dissociations suggesting these theories describe complementary functional layers rather than competing accounts. In Experiment 1, a no-rewire Self-Model lesion abolishes metacognitive calibration while preserving first-order task performance, yielding a synthetic blindsight analogue consistent with HOT predictions. In Experiment 2, workspace capacity proves causally necessary for information access: a complete workspace lesion produces qualitative collapse in access-related markers, while partial reductions show graded degradation, consistent with GWT's ignition framework. In Experiment 3, we uncover a broadcast-amplification effect: GWT-style broadcasting amplifies internal noise, creating extreme fragility. The B2 agent family is robust to the same latent perturbation; this robustness persists in a Self-Model-off / workspace-read control, cautioning against attributing the effect solely to z_{self} compression. We also report an explicit negative result: raw perturbational complexity (PCI-A) *decreases* under the workspace bottleneck, cautioning against naive transfer of IIT-adjacent proxies to engineered agents. A simple composite of markers (GBI + ΔPCI + AUROC) predicts OOD fragility. These results suggest a hierarchical design principle: GWT provides broadcast *capacity*, while HOT provides *quality control*. We emphasize that our agents are not conscious; they are reference implementations for testing functional predictions of consciousness theories.

1 Introduction

The science of consciousness faces a validation crisis. While theories like Global Workspace Theory (GWT) [Baars, 1988, Dehaene and Changeux, 2011], Integrated Information Theory (IIT) [Tononi, 2004, Oizumi et al., 2014], and Higher-Order Theories (HOT) [Rosenthal, 2005, Lau and Rosenthal, 2011] propose distinct neural signatures of awareness (ranging from non-linear "ignition" [Dehaene, 2014] and causal integration (Φ) [Oizumi et al., 2014] to metacognitive self-monitoring [Fleming et al., 2010]), testing them in biological systems is constrained by the inability to perform precise architectural ablations. We propose *synthetic neuro-phenomenology*: building agents that explicitly embody these theoretical commitments to test their functional utility.

This approach addresses a gap in both neuroscience and artificial intelligence. For neuroscience, synthetic agents offer a "perfect model organism" where every neuron, weight, and activation is observable and manipulable, allowing for causal tests that are ethically or technically impossible in biological brains. For AI, consciousness theories offer a potential roadmap for overcoming the brittleness of current systems [Bengio, 2017]. While deep learning has achieved broad success, modern models often lack the robustness, adaptability, and self-monitoring capabilities characteristic of biological intelligence. By implementing architectural priors derived from consciousness science (such as global broadcasting [Goyal et al., 2022] and self-modeling), we aim to determine whether these mechanisms confer the same functional advantages to artificial agents that they are hypothesized to provide to biological organisms.

However, prior attempts to model consciousness in silico have often fallen short of systematic validation. Many implementations rely on loose metaphorical translations of neural theories, where a simple feedback

arXiv：2512.19155v1 [cs.AI] 2025 年 12 月 22 日

我们能测试人工智能的意识理论吗？消融、标记与稳健性

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2025 年 12 月 23 日

摘要

对可靠意识指标的探索已分裂为竞争的理论阵营（全球工作空间理论（GWT）、综合信息理论（IIT）和高阶理论（HOT）），各自提出了不同的神经特征。我们采用合成神经现象学方法：构建具备这些机制的人工代理，通过生物系统无法实现的精确结构消融来测试其功能后果。在三项实验中，我们报告了分离现象，表明这些理论描述的是互补的功能层，而非相互竞争的叙述。在实验 1 中，一个无重接线的自模型病变在保留一阶任务表现的同时消除元认知校准，产生了与 HOT 预测一致的合成盲视类比。在实验 2 中，工作空间容量对信息获取具有因果性：完整的工作区病变会导致访问相关标志的质性崩溃，而部分减少则显示出渐变性退化，符合 GWT 点燃框架的特征。在实验 3 中，我们发现了一种广播放大效应：GWT 式广播会放大内部噪声，造成极度脆弱。B2 药物家族对相同的潜在扰动具有鲁棒性;这种稳健性在自建模关闭/工作区读取控制中依然存在，提醒人们不要将这种效果仅归因于 z 压缩。我们还报告了一个明确的负面结果：在工作区瓶颈下，原始扰动复杂度（PCI-A）下降，提醒人们不要将 IIT（印度理工学院）代理物简单地转移给工程化代理。简单合成的标志（GBI + ΔPCI + AUROC）可以预测 OOD 脆弱性。 这些结果表明采用层级设计原则：GWT 提供广播容量，HOT 负责质量控制。我们强调我们的代理人并非有意识;它们是用于测试意识理论功能预测的参考实现。

1 简介

意识科学正面临验证危机。而像全局工作空间理论（GWT）[Baars， 1988;Dehaene 和 Changeux， 2011]、整合信息理论（IIT）[Tononi， 2004，Oizumi 等， 2014]和高阶理论（HOT）[Rosenthal， 2005;Lau 和 Rosenthal，2011]提出了不同的神经意识特征（范围从非线性“点燃”[Dehaene， 2014]和因果整合（Φ）[Oizumi 等，2014]到元认知自我监控[Fleming 等， 2010 年），在生物系统中测试它们受限于无法进行精确的结构消融。我们提出综合神经现象学：构建明确体现这些理论承诺的代理，以测试其功能效用。

这种方法弥补了神经科学和人工智能领域的空白。对于神经科学来说，合成剂提供了一个“完美模式生物”，每个神经元、重量和激活都可以被观察和操控，从而实现生物大脑中伦理或技术上不可能的因果测试。对于人工智能来说，意识理论为克服当前系统脆弱性提供了潜在的路线图[Bengio， 2017]。尽管深度学习取得了广泛成功，现代模型往往缺乏生物智能特有的鲁棒性、适应性和自我监控能力。通过实施源自意识科学的架构先验（如全球广播[Goyal 等，2022]和自我建模），我们旨在确定这些机制是否赋予人工代理与它们被假设为生物体提供的相同功能优势。

然而，以往在计算机模拟中建模意识的尝试往往未能实现系统的验证。许多实现依赖于神经理论的松散隐喻翻译，其中只需简单的反馈

loop is labeled "recurrent processing" or a central variable is called a "global workspace" without establishing the functional necessity of these features. Furthermore, existing studies typically focus on single theories in isolation (building a "GWT agent" [Goyal et al., 2022] or a "HOT-inspired Self-Model agent" [Wilterson and Graziano, 2021]), ignoring the potential interactions and dependencies between these mechanisms. Few studies subject these architectures to "adversarial phenomenology": actively trying to break the system via lesions, noise, and deceptive signaling to distinguish genuine architectural advantages from mere task-specific overfitting.

In this work, we address these shortfalls by subjecting our agents to a battery of stress tests. We move beyond merely confirming theory predictions to stress-testing them in a controlled synthetic environment. To do this, we designed agents with reinforcement-learning-style architectures—trained here via supervised behavior cloning from an oracle (used only to generate training targets) to isolate architectural effects from optimization confounds—that incorporate specific architectural features proposed by consciousness theories. Specifically, we built agents with (1) a **global workspace** to broadcast information between modules (representing GWT), (2) a **Self-Model** to monitor internal states (representing HOT), and (3) **complexity probes** to measure information integration (inspired by IIT). Because these are software agents, we can perform manipulations that are impossible in biology: we apply **causal lesions** (selectively breaking parts of the "brain"), **noise titration** (injecting precise amounts of interference to test robustness), and **adversarial reward structures** (creating environments that incentivize the agent to "lie" to itself).

Our findings reveal a functional hierarchy rather than competing theories:

1. **Action Without Metacognition (Synthetic Blindsight):** Consistent with HOT predictions, we find that in this agent family, ablating the Self-Model selectively abolishes metacognitive calibration ("knowing that you know") while leaving first-order task performance largely intact, yielding a synthetic analogue of blindsight where agents can *do* without *knowing*.

2. **Bus-Off Discontinuity in Access Markers:** We observe an ignition-analogous discontinuity in this finite, discrete-capacity workspace: across the capacity settings we test, access markers remain similar across the tested non-zero capacities but collapse when the workspace is fully lesioned (bus-off)[1]. Reducing workspace capacity does not gradually dim access within the tested levels; instead, a complete workspace lesion produces a qualitative collapse in access-related markers.

3. **Broadcast-Amplification vs. Noise-Attenuation:** Contrary to the intuition that broadcasting creates robustness [Baars, 1988, Bengio, 2017], we find that a global workspace alone exhibits a *broadcast-amplification* effect: it broadcasts information but also amplifies noise, making the system fragile even under standard regularization and noise augmentation. In our implementation, the B2 agent family remains robust in the same perturbation regime (L75 > 0.50), including in a Self-Model-off / workspace-read control; we therefore treat noise-attenuation as an empirical robustness property of the B2 family rather than as a bottleneck-only effect.

4. **A Transfer Constraint on IIT-Adjacent Proxies:** Raw perturbational complexity (PCI-A) is *inverted* under the workspace bottleneck: the GWT agent shows lower trajectory complexity than baselines, contrary to a naive "more integrated $\Rightarrow$ higher PCI" expectation. This explicit negative result constrains how PCI-style measures transfer to engineered agents and suggests that absolute algorithmic complexity is not an architecture-invariant proxy for integration.

These results suggest that consciousness markers are not redundant. GWT provides the *capacity* for information exchange, while HOT mechanisms provide the *control* necessary to stabilize that exchange. Neither alone is sufficient for robust agency.

Scope and Limitations. Throughout this work, we focus strictly on *Access Consciousness* (information availability for report and action) rather than phenomenal consciousness or subjective experience. Our agents are not conscious; they are reference implementations of proposed mechanisms. Our claims concern functional correlates within these synthetic systems: whether certain architectural features are necessary or sufficient for particular behavioral and representational signatures. We make no claims about whether these features would produce phenomenal experience in biological or artificial systems. Because we evaluate up to $n = 20$ seeds per condition (Experiments 1–3; seed as the unit of analysis), our statistical claims are best viewed as proof-of-concept evidence; we highlight large qualitative effects and report uncertainty estimates for transparency. Some targeted sweeps and attribution controls are pilots ($n = 5$ seeds) and are labeled as such.

[1] Here, we use "ignition-analogous" in a loose access sense: an abrupt change between tested *discrete* capacity settings in a finite system. We do *not* claim a thermodynamic-limit phase transition, power-law scaling, or a precisely estimated critical point.

循环被称为“循环处理”，或者一个中心变量被称为“全局工作区”，但并未确立这些功能的功能必要性。此外，现有研究通常只关注单一理论（构建“GWT 代理”[Goyal 等，2022]或“受 HOT 启发的自模型代理”[Wilterson 和 Graziano，2021]），忽视了这些机制之间潜在的相互作用和依赖关系。很少有研究对这些架构进行“对抗现象学”研究：通过病变、噪声和欺骗信号主动破坏系统，以区分真正的架构优势与单纯的任务特定过拟合。

在本研究中，我们通过对代理人进行一系列压力测试来解决这些不足。我们不仅仅是验证理论预测，而是在受控的合成环境中进行压力测试。为此，我们设计了具有强化学习风格架构的代理——这里通过预言机的监督行为克隆训练（仅用于生成训练目标），以隔离架构效应与优化混杂因素——这些代理融合了意识理论提出的具体架构特征。具体来说，我们构建了具备（1）全局工作区用于模块间信息广播（代表 GWT）、（2）自模型监控内部状态（代表 HOT）、以及（3）复杂度探针用于测量信息整合度（受 IIT 启发）的代理。由于这些是软件代理，我们可以执行生物学中不可能的操作：我们施加因果损伤（选择性破坏“大脑”部分）、噪声滴定（注入精确干扰以测试鲁棒性）和对抗性奖励结构（创造激励代理“欺骗”自身的环境）。

我们的发现揭示了一个功能层级结构，而非竞争理论：

1. 无元认知的行动（合成盲视）：与 HOT 预测一致，我们发现在该代理家族中，选择性地消除自我模型会消除元认知校准（“知道你知道”），同时基本保留一阶任务表现，形成一种合成的盲视类比，代理可以不知情地完成任务。

2. 接入标志的总线断裂：我们观察到该有限离散容量工作空间中类似点火的不连续性：在测试的容量设置中，接入标志在测试的非零容量内保持相似，但当工作空间完全受损（总线关闭）时会坍塌。减少工作空间容量并不会逐渐降低测试水平内的可达性;相反，完整的工作区病变会导致通路相关标志物的质性崩溃。

3. 广播放大与噪声衰减：与广播带来鲁棒性的直觉相反 [Baars， 1988;Bengio， 2017]，我们发现仅全球工作空间就表现出广播放大效应：它广播信息，同时放大噪声，使系统即使在标准正则化和噪声增强下也脆弱。在我们的实现中，B2 代理家族在相同的扰动区间（L75 > 0.50）中保持稳健，包括自建模/工作区读取控制;因此，我们将噪声衰减视为 B2 家族的经验鲁棒性特性，而非仅限瓶颈效应。

4. IIT 邻接代理的传输约束：原始扰动复杂度（PCI-A）在工作区瓶颈下被反转：GWT 代理显示出比基线更低的轨迹复杂度，这与天真的“更集成⇒更高 PCI”预期相反。这一明确的负结果限制了 PCI 风格度量如何传递给工程代理，并表明绝对算法复杂度并非架构不变的集成代理。

这些结果表明意识标记并非多余。GWT 提供信息交换的能力，而 HOT 机制则提供稳定该交换所需的控制。

仅凭这两者都不足以构成强有力的能动性。

范围与限制。在整个工作中，我们严格关注访问意识（报告和行动所需的信息可用性），而非现象意识或主观体验。我们的代理人是没有意识的;它们是拟议机制的参考实现。我们的主张关注这些综合系统内的功能相关性：某些架构特征是否对特定行为和表征特征是必要或充分的。我们不对这些特征是否能在生物或人工系统中产生惊人的体验做出断言。由于我们每个条件最多评估 n = 20 个种子（实验 1–3;以种子为分析单位），我们的统计主张最好被视为概念验证证据;我们强调了大型定性影响，并报告不确定性估计以实现透明度。一些有针对性的扫除和归因控制是试点。许多实现依赖于神经理论的松散隐喻翻译（n=5 种子），并以此为标签。

[1] 这里，我们用“点火类比”来指宽松的访问：在有限系统中，经过测试的离散容量设置之间的突然变化。我们不主张热力学极限相变、幂律尺度或精确估计的临界点。

2 Background

2.1 Neuroscience & Consciousness Concepts

To bridge the gap between biological theory and synthetic implementation, we clarify key neuroscientific terms used throughout this work.

Type-1 vs. Type-2 Performance. In psychophysics, **Type-1 performance** refers to the objective accuracy of a subject's decision (e.g., "is the stimulus left or right?"). **Type-2 performance** refers to the subject's ability to monitor their own accuracy, typically measured via confidence ratings (e.g., "how sure are you?"). A key signature of conscious awareness is the correlation between Type-1 and Type-2 performance (metacognitive sensitivity). Conditions like *blindsight* [Weiskrantz, 1986] demonstrate that high Type-1 performance can exist without Type-2 awareness, a dissociation we model in Experiment 1.

Ignition and Broadcasting. Global Workspace Theory posits that conscious access involves a non-linear "ignition" event, an abrupt thresholding dynamic in which a stimulus overcomes a barrier to trigger sustained, widespread reverberation across fronto-parietal networks [Dehaene et al., 2006]. This "broadcasting" makes the information available to other cognitive processes. We model the capacity-limited broadcast aspect via a bottlenecked bus architecture; our implementation tests capacity constraints on information maintenance and broadcast, rather than competitive access gating.

Backward Masking and Visual Working Memory. *Backward masking* is an experimental technique in which a brief target stimulus is followed by a "mask" (a second stimulus) that interferes with conscious perception of the target. The *stimulus-onset asynchrony* (SOA), the time between target and mask onset, determines whether the target reaches awareness. This paradigm probes the temporal dynamics of conscious access. *Visual working memory* refers to the capacity-limited system that maintains visual information over short delays (typically 3–4 items in humans [Luck and Vogel, 1997, Cowan, 2001]). We use masking-like interference and capacity manipulations to test analogous dynamics in our synthetic agents.

Wagering Paradigms. A *wagering* or *post-decision wagering* paradigm asks subjects to bet on the correctness of their own decisions [Persaud et al., 2007]. High wagers on correct trials and low wagers on incorrect trials indicate good metacognitive calibration. An *opt-out* option (declining to wager) provides an additional measure of subjective uncertainty. We adapt this paradigm to assess whether our agents can calibrate confidence to their own accuracy.

TMS–EEG and PCI. The Perturbational Complexity Index (PCI) [Casali et al., 2013] is measured by perturbing the brain with transcranial magnetic stimulation (TMS, a non-invasive technique that induces neural activity via magnetic pulses) and recording the resulting spatiotemporal dynamics with electroencephalography (EEG, scalp recordings of electrical brain activity). PCI quantifies the algorithmic complexity of the brain's response: conscious states produce complex, differentiated responses, while unconscious states (e.g., deep sleep, anesthesia) produce simpler, more stereotyped responses. Our PCI-A proxy applies an analogous perturbation-and-compress logic to synthetic agent trajectories.

2.2 Related Work

Our work builds on and extends several research threads that have previously remained largely separate.

Consciousness-Inspired AI Architectures. Several recent efforts have sought to implement consciousness-related mechanisms in artificial systems. Goyal et al. [2022] introduced recurrent independent mechanisms with attention-based communication, inspired by GWT's broadcast architecture. VanRullen et al. [2021] explored attention-gated global workspaces in vision models. Wilterson and Graziano [2021] developed HOT-inspired architectures with explicit self-monitoring. However, these works typically focus on single theories in isolation and demonstrate performance gains without systematic ablation or stress-testing to establish causal necessity. Our contribution is to (1) implement multiple theoretical mechanisms within a unified architecture, (2) systematically lesion each component to establish necessity, and (3) subject the system to adversarial perturbations that distinguish genuine robustness from task-specific fitting.

2 背景

2.1 神经科学与意识概念

为了弥合生物学理论与合成实现之间的鸿沟，我们澄清了本研究中常用的关键神经科学术语。

Type-1 与 Type-2 性能。在心理物理学中，类型 1 表现指的是受试者决策的客观准确性（例如，“刺激是左还是右？”）。2 型表现指的是受试者监控自身准确性的能力，通常通过置信度评级（例如，“你有多确定？”）来衡量。意识意识的一个关键特征是第一型与第二型表现之间的相关性（元认知敏感性）。盲视等条件[Weiskrantz， 1986]表明，高一型表现可以在没有二型意识的情况下存在，而我们在实验 1 中模拟了这种解离。

点火与广播。全球工作空间理论认为，有意识的访问涉及非线性“点燃”事件，即刺激突破障碍，触发额骨-顶叶网络持续且广泛的回响的突然阈值动态[Dehaene 等，2006]。这种“广播”使信息能够被其他认知过程调用。我们通过瓶颈总线架构来建模容量有限的广播部分;我们的实现测试信息维护和广播的能力限制，而非竞争性访问门控。

反向掩蔽与视觉工作记忆。反向掩蔽是一种实验技术，在短暂的目标刺激之后，接着一个“掩蔽”（第二个刺激），干扰对目标的意识感知。刺激-启动非同步（SOA），即目标与掩体启动之间的时间，决定目标是否达到意识状态。该范式探讨了意识访问的时间动态。视觉工作记忆指的是容量有限的系统，该系统在短时间内（人类通常为 3–4 项）维持视觉信息 [Luck 和 Vogel， 1997;Cowan， 2001]。我们使用类似掩蔽的干扰和容量操作来测试合成代理中的类似动态。

投注范式。投注或决策后投注范式要求受试者对自己决策的正确性进行投注[Persaud 等，2007]。对正确试验的高额投注和错误试验的低赌注表明元认知校准良好。选择退出选项（拒绝下注）则增加了额外的主观不确定性。我们调整这一范式，以评估我们的代理是否能够根据自身准确性校准置信度。

TMS–EEG 和 PCI。扰动复杂性指数（PCI）[Casali 等，2013]通过经颅磁刺激（TMS，一种通过磁脉冲诱导神经活动的非侵入技术）扰动大脑，并用脑电图（EEG，头皮脑电活动记录）记录由此产生的时空动态来测量。PCI 量化了大脑反应的算法复杂性：意识状态产生复杂且有差异化的反应，而无意识状态（如深度睡眠、麻醉）则产生更简单、更刻板的反应。我们的 PCI-A 代理对合成代理轨迹应用类似的扰动与压缩逻辑。

2.2 相关工作

我们的工作建立在并扩展了几个此前大多独立的研究线索之上。

意识启发的人工智能架构。近期有若干努力试图在人工系统中实现与意识相关的机制。Goyal 等人 [2022] 引入了基于注意力的循环独立沟通机制，灵感来自 GWT 的广播架构。VanRullen 等人 [2021] 探索了视觉模型中的注意力门控全球工作空间。Wilterson 和 Graziano [2021] 开发了受 HOT 启发的带有显式自我监控的架构。然而，这些研究通常聚焦于孤立的单一理论，并在没有系统消融或压力测试以确立因果必然性的情况下，展示性能提升。我们的贡献是：（1）在统一架构中实现多种理论机制，（2）系统地分解每个组件以确立必要性，（3）使系统受到对抗扰动，从而区分真正的鲁棒性与任务特定拟合。

Metacognition and Calibration in Machine Learning. The machine learning literature has extensively studied uncertainty quantification and calibrated confidence [Guo et al., 2017]. Selective prediction frameworks [Geifman and El-Yaniv, 2017] allow models to abstain when uncertain, and recent work connects these ideas to human-like metacognition [Raileanu and Fergus, 2020]. Our Type-2 AUROC metric derives from this literature but is deployed here to test HOT-specific predictions: that metacognitive calibration requires a higher-order self-representation rather than emerging automatically from first-order competence. The synthetic blindsight dissociation we report (preserved task performance with abolished metacognition) provides a controlled test case for these ideas.

Robustness and Noise in Neural Networks. The fragility of deep networks to adversarial perturbations [Szegedy et al., 2014] and distribution shift [Hendrycks and Dietterich, 2019] is well documented. Prior work has explored noise injection during training for regularization [Srivastava et al., 2014] and studied the geometry of learned representations under perturbation [Mahloujifar et al., 2019]. Our broadcast-amplification finding connects to this literature but focuses on *latent* perturbations (noise in the workspace, not inputs) and demonstrates that broadcasting alone amplifies rather than filters noise. The B2 family's robustness in the same latent-noise regime suggests that additional architectural structure and training objectives can provide complementary robustness mechanisms beyond input-level defenses.

Metacognition in Reinforcement Learning. While uncertainty quantification in deep learning is well-established [Guo et al., 2017], recent work has begun to explore the functional role of metacognition in reinforcement learning agents. Anderson and Oates [2020] demonstrated that metacognitive monitoring can enhance perturbation tolerance in RL agents, a finding consistent with our noise-attenuation hypothesis. Similarly, Salem-Garcia et al. [2023] linked confidence biases to fundamental RL processes, suggesting that miscalibration is not merely a post-hoc artifact but can actively shape learning dynamics. Our work extends these perspectives by causally dissociating first-order performance from second-order confidence via architectural lesions, providing a synthetic testbed for theories that posit hierarchical relationships between task competence and metacognitive awareness [Fleming, 2024].

3 From Theories to Synthetic Indicators

Having established the relevant neuroscientific terminology and experimental paradigms, we now turn to the core challenge of this work: translation. To move beyond loose analogies between brains and machines, we must operationalize theoretical constructs into precise, measurable indicators within our synthetic agents. We selected three dominant clusters of consciousness theories (HOT, GWT, and IIT) because they address distinct functional facets of consciousness that are often conflated: (1) **Monitoring** (reflexive awareness), (2) **Access** (information availability), and (3) **Integration** (phenomenological unity). By testing these pillars individually, we can determine whether they are redundant or complementary.

Higher-Order Theories (HOT). HOT suggests that a state is conscious only when the system represents itself as being in that state [Rosenthal, 2005, Lau and Rosenthal, 2011]. This corresponds to the **Monitoring** dimension: access alone is insufficient for confidence, error correction, and a "personal" perspective. We implement this via a dedicated **Self-Model** (z_{self}) that compresses the workspace state into a summary latent. The indicator is **Metacognitive Calibration** (Type-2 AUROC), measuring the system's ability to "know that it knows" [Fleming et al., 2010]. We test this in Experiment 1 to dissociate *doing* (first-order performance) from *knowing* (second-order awareness).

Global Workspace Theory (GWT). GWT posits that consciousness arises from a global broadcast mechanism where information from specialized modules is selected and amplified ("ignited") for system-wide access [Baars, 1988, Dehaene and Changeux, 2011]. This corresponds to the **Access** dimension: without global broadcasting, there is no "content" to be conscious of. In our agents, we operationalize this as a bottlenecked bus architecture. The key indicators are **Ignition**, a sudden, non-linear jump in workspace activation, and the **Global Broadcast Index (GBI)**, quantifying the extent of information sharing across the system. We test this in Experiment 2 to establish the causal substrate of information access.

机器学习中的元认知与校准。机器学习文献对不确定性量化和校准置信度进行了广泛研究 [Guo 等，2017]。选择性预测框架[Geifman 和 El-Yaniv， 2017]允许模型在不确定时保持禁欲，近期研究将这些理念与类人类元认知联系起来[Raileanu 和 Fergus， 2020]。我们的 2 型 AUROC 指标源自这些文献，但用于测试 HOT 特有的预测：元认知校准需要更高阶的自我表征，而非一阶能力自动产生。我们报告的合成盲视解离（保留任务表现但废除元认知）为这些观点提供了一个受控的测试案例。

《神经网络中的鲁棒性与噪声》。深度网络对对抗扰动的脆弱性[Szegedy 等，2014]和分布转移[Hendrycks 和 Dietterich，2019]的脆弱性已有充分文献记载。此前的研究曾探索正则化训练中的噪声注入[Srivastava 等，2014]，并研究了微扰下学习表现的几何[Mahloujifar 等，2019]。我们的广播放大发现与相关文献相呼应，但侧重于潜在扰动（工作空间噪声，而非输入），并证明广播本身放大噪声，而非过滤。B2 家族在同一潜噪声环境中的鲁棒性表明，额外的架构结构和训练目标可以提供超越输入级防御的互补鲁棒机制。

强化学习中的元认知。虽然深度学习中的不确定性量化已被广泛认可[Guo 等，2017]，但近期研究已开始探索元认知在强化学习代理中的功能作用。Anderson 和 Oates [2020] 证明元认知监测能增强强化学习代理的干扰容忍度，这一发现与我们的噪声衰减假说一致。同样，Salem-Garcia 等人[2023]将置信偏差与基本强化学习过程联系起来，表明校准错误不仅是事后产物，还能积极影响学习动态。我们的工作通过结构损伤因果地将一阶性能与二阶信心分离，扩展了这些视角，为假设任务能力与元认知意识之间存在层级关系的理论提供了综合测试平台[Fleming，2024]。

3　从理论到综合指标

在建立了相关的神经科学术语和实验范式后，我们现在转向本工作的核心挑战：翻译。为了超越大脑与机器之间的松散类比，我们必须将理论构造操作化为合成代理内精确且可测量的指标。我们选择了三大主导意识理论簇（HOT、GWT 和 IIT），因为它们涉及意识中常被混淆的不同功能方面：（1）监测（反思觉知），（2）获取（信息可用性），（3）整合（现象学统一性）。通过单独测试这些支柱，我们可以判断它们是重复还是互补。

高阶理论（HOT）。HOT 认为，只有当系统将自己表现为处于该状态时，状态才是有意识的[Rosenthal， 2005;Lau 和 Rosenthal， 2011]。这对应于监控维度：仅凭访问不足以实现信心、纠错和“个人”视角。我们通过专用的自模型（z）实现，将工作区状态压缩为摘要潜在状态。该指标是元认知校准（2 型 AUROC），衡量系统“知道它知道”的能力[Fleming 等，2010]。我们在实验 1 中测试此点，以将“做”（一阶表现）与“知道”（二阶意识）分离。

全球工作空间理论（GWT）。GWT 认为，意识源自一种全球广播机制，其中从专业模块中选择并放大（“点燃”）信息以实现系统范围的访问[Baars， 1988， Dehaene and Changeux， 2011]。这对应于接入维度：没有全球广播，就没有需要被意识到的“内容”。在我们的代理中，我们将此作为瓶颈总线架构来操作化。关键指标是点火（Ignition），即工作空间激活的突然非线性跳跃，以及全球广播指数（GBI），用于量化系统间信息共享的程度。我们在实验 2 中测试此点，以确定信息获取的因果基础。

Integrated Information Theory (IIT). IIT argues that consciousness corresponds to the capacity of a system to integrate information, measured as Φ [Tononi, 2004, Oizumi et al., 2014]. This corresponds to the **Integration** dimension: consciousness is characteristically unified and differentiated.

We do not compute Φ directly. Instead, we include a PCI-inspired perturbational complexity probe [Casali et al., 2013] as a *transfer test*: if "PCI-like" trajectory complexity were a broadly architecture-invariant proxy for IIT-style integration, then architectures that enforce more global coupling and causal interdependence should tend to exhibit higher perturbation-evoked complexity.

However, in engineered agents this expectation is not guaranteed [Shwartz-Ziv and Tishby, 2017, Ansuini et al., 2019]. In particular, a global workspace can behave as a *bottleneck* and *organizer*: by forcing many downstream computations to depend on a shared, low-rank broadcast state, it can increase redundancy and push dynamics onto a lower-dimensional manifold. Under such conditions, stronger architectural "integration" (in the everyday engineering sense of tighter coupling) can plausibly produce *lower* Lempel–Ziv compressibility-based trajectory complexity.

Accordingly, we treat raw PCI-A as a falsifiable probe rather than a confirmatory marker, and we interpret deviations (including inversions) as informative constraints on how PCI-style measures do or do not transfer to artificial systems. We report both raw PCI-A (absolute post-perturbation trajectory complexity) and a contrastive measure, **ΔPCI** (correct minus incorrect trial complexity), which aims to partially control for the fact that in these agents, "complexity" can be inflated by unstructured failure/noise. We include this in Experiment 3 to test which aspects of PCI-style proxies behave monotonically with architectural changes in this synthetic setting.

These indicators form a distinct profile for each theory, summarized in Table 1.

Table 1: Mapping theoretical clusters to synthetic indicators and experiments.

Exp.	**Theory Cluster**	**Intended Property**	**Synthetic Indicator**
E1	HOT / Self-Model	Metacognitive access	Type-2 AUROC, z_{self} lesion
E2	GWT	Global access, ignition	GBI, Ignition Sharpness
E3	Complexity probe (IIT-adjacent)	Richness, differentiation	PCI-A (raw), ΔPCI
E3	Composite	Joint "conscious-like" profile	Reduced composite (GBI + ΔPCI); full CTS adds AUROC

4 Methods

4.1 Architectures

We compared a hierarchy of agent architectures, all sharing a common backbone but differing in their high-level cognitive modules.

Common Backbone. All agents use a convolutional encoder (3-layer CNN) to process the $\mathbf{O}_{\text{grid}} \in \mathbb{N}^{7\times7\times3}$ input, producing a 151-dimensional observation embedding.

Feedforward Baseline (A0). A purely reactive agent consisting of an MLP (Multi-Layer Perceptron) policy head attached directly to the visual encoder. It lacks memory or internal state, serving as a control for tasks requiring temporal integration.

Recurrent Baseline (A1). A standard recurrent agent where the observation embedding feeds into a Gated Recurrent Unit (GRU) with 64 hidden units. The policy head reads directly from the GRU hidden state. This represents the "status quo" of deep reinforcement learning agents—capable of memory but lacking explicit workspace or Self-Model structures.

GWT Agent (B1): The Global Workspace. This agent implements a capacity-limited broadcast mechanism.

- **Workspace Bus:** A set of K memory slots (default $K = 4$, dimension $D = 16$).

综合信息理论（IIT）。IIT 认为，意识对应于系统整合信息的能力，测量为Φ[Tononi， 2004;Oizumi et al.， 2014]。这对应于整合维度：意识具有特征性地统一且有差异化。

我们不直接计算Φ。相反，我们包含一个受 PCI 启发的微扰复杂度探针[Casali 等，2013]作为转移测试：如果“类 PCI”轨迹复杂度是 IIT 式集成的广泛架构不变代理，那么强制执行更多全局耦合和因果相互依赖的架构应表现出更高的扰动诱发复杂度。

然而，在工程化代理中，这种期望并不保证 [Shwartz-Ziv 和 Tishby， 2017， Ansuini 等， 2019]。特别是，全局工作区可以作为瓶颈和组织者：通过迫使许多下游计算依赖于共享的低秩广播状态，可以增加冗余性，并将动态推向更低维的流形。在这种情况下，更强的架构“积分”（在日常工程意义上的更紧密耦合）可以合理地降低基于 Lempel–Ziv 可压缩性的轨迹复杂度。

因此，我们将原始 PCI-A 视为可证伪的探针而非确认标记，并将偏差（包括反转）解释为关于 PCI 风格度量如何转移至人工系统的有用约束。我们报告了原始 PCI-A（绝对扰动后轨迹复杂度）和对比度量量ΔPCI（正确减去错误试验复杂度），旨在部分控制这些因子中“复杂性”可能被非结构化失效/噪声膨胀的事实。我们将此纳入实验 3，以测试 PCI 风格代理在该合成环境中与架构变化的单调行为。

这些指标为每个理论构成了独特的轮廓，详见表 1。

表 1：将理论集群映射到综合指标和实验。

实验理论簇	预定财产	合成指示器
E1 HOT / 自建模元认知访问 类型 2 AUROC，ZLE-		
		西恩
E2 GWT 全局访问，点火 GBI，点火锐利度 E3 复杂度探针（IIT 邻近）	丰富度，微分 PCI-A（RAW），ΔPCI	
E3 复合		
	关节“有意识”剖面约化复合体（GBI +	
	ΔPCI）;完整 CTS 新增 AUROC	

4 方法

4.1 架构

我们比较了一个智能体架构的层级结构，它们共享共同的骨干，但在高层认知模块上有所不同。

共同脊梁。所有智能体都使用卷积编码器（三层 CNN）处理 O∈N 输入，产生 151 维的观测嵌入。

前馈基线（A0）。一个纯反应式代理，由直接连接到视觉编码器的 MLP（多层感知器）策略头组成。它缺乏内存或内部状态，作为需要时间整合任务的控制。

复发基线（A1）。一种标准的循环性智能体，其中观察嵌入输入一个包含 64 个隐藏单元的门控循环单元（GRU）。政策负责人直接从 GRU 隐藏状态中朗读。这代表了深度强化学习代理的“现状”——能够记忆，但缺乏显式的工作空间或自我建模结构。

GWT 代理（B1）：全球工作空间。该代理实现了容量有限的广播机制。

- 工作区总线：一组 K 个内存槽（默认 K = 4，维度 D = 16）。

- **Write Mechanism:** Recurrent state features are projected to a write vector and stored in the fixed-capacity buffer. At each step, the write vector is placed into the first available empty slot; if all slots are occupied, a fixed slot (slot 0) is overwritten.[2] This deterministic allocation tests *capacity limitation* and *broadcast*, not competitive attention-based access gating.

- **Broadcast (Read):** The content of all slots is aggregated and broadcast back to the policy head. We use a flattened-readout MLP that concatenates all workspace slots into a single vector and projects through a multi-layer network to produce action logits.

- **Capacity Control:** We can manipulate the number of active slots K (full capacity $K = 4$, intermediate reductions, and $K = 0$ (bus-off) for a complete lesion).

Self-Model Agent (B2; HOT-inspired). This agent extends B1 by adding a higher-order monitoring system.

- **Self-Model (z_{self}):** A secondary module that does not directly see the world. Instead, it observes the *workspace broadcast* (a D-dim summary), the *recurrent carrier state*, and the *agent's own intended actions* (policy logits).

- **Compression:** It compresses this high-dimensional input (89 dims: 64-d recurrent carrier + 16-d workspace broadcast + 7 action logits + 2 uncertainty scalars) into a tiny latent bottleneck (z_{self}, dim=64, effectively lower due to regularization).

- **Policy Routing:** The confidence head (for wagering) reads *exclusively* from z_{self}. The action policy routing varies by experiment: in Experiment 1 (metacognition), the action policy reads directly from the workspace (not z_{self}) so that Self-Model lesions preserve first-order performance while abolishing metacognition. In Experiment 3 (noise robustness), we evaluate both configurations: a "workspace-read" control where the action policy bypasses z_{self}, and the standard B2 where the action policy reads from z_{self}.

Baseline Variants for Disentangling Contributions. Two controls help separate the roles of broadcast and self-modeling. (i) We evaluate B2 with the Self-Model enabled under a complete workspace lesion (bus-off; $K = 0$), which tests whether a monitor can produce spurious metacognition when its monitored substrate is absent; Type-2 AUROC collapses to chance (see Experiment 1). (ii) We include compression and pathway controls: blind ablations (noise/permutation) and a PCA baseline matched to the effective dimensionality of z_{self} to show that metacognitive information depends on the learned content of z_{self}, not merely on the presence of a bottleneck.

Denoising attribution baselines (Experiment 3 controls). To tighten attribution for the robustness effects in Experiment 3, we additionally evaluate two baseline families. First, a "HOT-only" agent (A1 + z_{self} without any workspace bus) tests whether a Self-Model bottleneck can improve robustness in the absence of a broadcast substrate. Because this agent has no workspace bus, it cannot be evaluated under the strong-lesion cue routing (cues-to-workspace-only), so cues remain visible in its observation stream; we probe robustness by injecting noise into the recurrent carrier state (post-GRU hidden state) rather than into workspace slots. Second, we include matched "BC-only" compressor controls that replace z_{self} with a 64-d bottleneck applied directly to the flattened workspace state (linear projection, small MLP, and fixed random projection). These controls distinguish a Self-Model-specific robustness gain from a generic benefit of adding a low-dimensional bottleneck; results are summarized in Figure 4.

4.2 Environments

All experiments were conducted within customized **MiniGrid** environments [Chevalier-Boisvert et al., 2018], a standard 2D gridworld framework where agents perceive the world through a local view (7×7 grid) and take discrete actions (move forward, turn left, turn right, pick up, drop, toggle, done). The observation space consists of a symbolic tensor $\mathbf{O}_{\text{grid}} \in \mathbb{N}^{7\times7\times3}$ (object ID, color, state) plus a task-specific vector.

[2]We use fixed-slot overwrite rather than true FIFO (evicting the oldest entry) for implementation simplicity. In our short-episode tasks (32 steps) with sparse cue-write events, the workspace rarely fills completely, so the fallback policy seldom activates. When it does, the flattened-readout architecture ensures all slot contents remain accessible to the policy regardless of which slot was overwritten. This design choice therefore does not affect the validity of our capacity-limitation experiments.

・ 写入机制：循环状态特征投影为写向量并存储在固定容量缓冲区中。每一步，写入向量被放入第一个可用的空槽;如果所有槽位都被占用，则覆盖一个固定槽位（槽位 0）。这种确定性分配测试的是容量限制和广播，而非基于注意力的竞争性访问门控。

- 广播（读取）：所有时隙的内容被汇总并广播回给政策主管。我们使用一个扁平读出 MLP，将所有工作区槽位串接成单一矢量，并通过多层网络进行项目生成动作日志。

- 容量控制：我们可以操作活动槽数 K（满容量 K=4，中间减小，K=0（总线关闭）以实现完全病变）。

自建模代理（B2;受热点启发）。该智能体通过添加高阶监控系统扩展了 B1。

- 自建模（z）：一个不直接观察世界的次要模块。相反，它观察工作区广播（D-dim 摘要）、重复载波状态以及代理自身的预期操作（策略日志）。

- 压缩：它将高维输入（89 个 DIMS：64 维递归载波+16 维工作区广播+7 个动作对数+2 个不确定性标量）压缩到一个极小的潜在瓶颈（z，dim=64，由于正则化而实际上更低）。

- 策略路由：信心中心（用于投注）仅从 z 开始。动作策略的路由因实验而异：在实验 1（元认知）中，动作策略直接读取工作空间（而非 z），使自模型病灶保持一阶性能，同时消除元认知。在实验 3（噪声鲁棒性）中，我们评估了两种配置：一个是“工作区读”控制，其中动作策略绕过了 z;另一个是标准 B2，即动作策略从 z 读取。

解开纠缠贡献的基线变体。两个控制区分广播和自建模的角色。（i）在完整工作区病变（总线切除）下，我们评估启用自模型的 B2;K = 0），该测试在监测基底缺失时，监测者是否能产生虚假的元认知;2 型 AUROC 因偶然崩塌（见实验 1）。（ii） 我们纳入了压缩和通路控制：盲测消融（噪声/置换）以及与 zto 有效维度匹配的 PCA 基线表明，元认知信息依赖于 z 的学习内容，而不仅仅是瓶颈的存在。

去噪归因基线（实验 3 对照组）。为了严格归属
在实验 3 中，我们还评估了两个基线家族。首先，“仅 HOT”代理（A1 + z，无任何工作区总线）测试自建模瓶颈是否能在无广播基底下提升鲁棒性。由于该代理没有工作区总线，无法通过强病灶线索路由（仅线索到工作区）进行评估，因此线索仍可在观察流中可见;我们通过向递归载波态（后 GRU 隐藏态）注入噪声，而非工作区槽来探测鲁棒性。其次，我们包含匹配的“仅限 BC”压缩器控制，用直接应用于平面工作区状态的 64 维瓶颈替代了 z（线性投影、小 MLP 和固定随机投影）。这些控制区分了自模型特异性的鲁棒性收益与添加低维瓶颈的通用优势;结果总结见图 4。

4.2 环境

所有实验均在定制的 MiniGrid 环境内进行[Chevalier-Boisvert 等，2018]，这是一种标准的二维网格世界框架，代理通过局部视角（7×7 网格）感知世界，并采取离散动作（前进、左转、右转、拾取、放下、切换、完成）。观测空间由符号张量 O∈N（对象 ID、颜色、状态）加上一个任务特定的向量组成。

[3] 我们采用固定槽覆盖，而不是真正的先出（FIFO，淘汰最早条目）以简化实现。在我们短集任务（32 步）中，提示写入事件稀疏，工作区很少会完全填满，因此备份策略很少启动。当它被覆盖时，平面读出架构确保所有槽内容无论哪个槽被覆盖，策略都能访问。因此，这一设计选择不影响我们容量限制实验的有效性。

Wagering Paradigm (Experiment 1). To assess metacognition, we implemented a wagering task within the MiniGrid framework. The agent navigates to a "stimulus zone" where a transient cue (encoded via the standard MiniGrid observation channels) appears with variable salience (simulating perceptual uncertainty). The agent must then navigate to one of two "response zones" to indicate its classification.

- **Task:** Classify which of two cue types ($s \in \{0, 1\}$) was presented, where cue salience is degraded by observation noise.
- **Wager:** On 50% of trials, a "wager flag" appears after the decision, prompting the agent to output a continuous confidence signal $c \in [0, 1]$ via a dedicated action channel.
- **Reward Structure:** High confidence ($c > 0.5$) counts as a "bet" (+1 reward for correct, -1 for incorrect). Low confidence ($c \leq 0.5$) counts as an "opt-out" (0 reward). This incentivizes the agent to align its internal confidence c with its objective probability of correctness $P(\text{correct})$.

Dual-Task Paradigm (Experiment 2). To test GWT predictions, we used a `MiniGrid-Empty-5x5-v0` variant with a dual-task structure. Each episode (max 32 steps) involves:

1. **Encoding Phase:** Agents observe two transient cues: a "secondary" cue (color) and a "primary" cue (object type).
2. **Delay Phase:** Agents must navigate the empty room while maintaining both cues in working memory.
3. **Report Phase 1 (Early):** A specific "report flag" signal triggers the agent to output the secondary cue.
4. **Report Phase 2 (Late):** A second flag triggers the agent to output the primary cue.

We employed a "strong-lesion" wiring: cue information was routed *exclusively* through the workspace bus, while the standard recurrent encoder was blind to these specific cue channels. This ensures the workspace is the sole causal pathway for task-relevant information.

Latent Noise Stress Test (Experiment 3). To evaluate robustness on the dual-task paradigm (Experiment 2), we introduced a Latent Workspace Perturbation protocol. Instead of adding noise to the pixel observations (which traditional CNNs can easily filter), we injected Gaussian noise $\mathcal{N}(0, \sigma^2)$ directly into the agent's internal workspace slots immediately *after* the write operation but *before* the read operation. For recurrence-only controls without a workspace (HOT-only), we analogously inject Gaussian noise into the recurrent carrier state (post-GRU hidden state) immediately after the recurrent update and before downstream readouts. This "brain stimulation" approach tests the system's stability against internal degradation. We measured the **L75**: the noise level σ at which report-step decision accuracy drops to 75%.[3] This effectively quantifies the "tolerance margin" of the cognitive architecture.

4.3 Training

Agents were trained using **Behavior Cloning (BC)** from an expert oracle to ensure baseline competence. We use BC as an *optimization control*: it reduces RL-specific confounds (exploration failures, reward sparsity, and training instability) so that differences under lesions and perturbations can be attributed to architectural manipulations rather than optimization dynamics. The oracle is used *only* to generate action targets during training; it is never consulted at inference. All evaluations are autonomous rollouts of the learned policy, and our central results are established via *post-training* causal interventions (lesions, capacity ablations, and latent workspace noise) applied to trained agents. This framing makes our claims conditional: given agents that reach baseline first-order competence, which components are causally necessary for access- and metacognition-related signatures?

[3] We use "L75" rather than "LD50" to avoid confusion with the toxicology convention where LD50 denotes the dose causing 50% lethality. Our threshold uses 75% accuracy (not 50% accuracy).

投注范式（实验 1）。为了评估元认知，我们在 MiniGrid 框架内实施了一个投注任务。代理会导航到一个“刺激区”，在那里出现一个瞬态线索（通过标准 MiniGrid 观察通道编码），显著性可变（模拟感知不确定性）。代理人随后必须导航到两个“响应区”之一以显示其分类。

- 任务：分类两种提示类型（$s \in \{0, 1\}$）中哪一种，其中提示显著性会被观察噪声削弱。

- 下注：在 50%的试验中，决策后会出现“下注标志”，提示代理通过专用动作通道输出连续置信信号 $c \in [0, 1]$。
- 奖励结构：高信心（$c > 0.5$）算作“投注”（正确+1 奖励，错误减 1）。低信心（$c \leqslant 0.5$）被视为“选择退出”（0 奖励）。这激励代理人将其内部置信度 c 与其客观正确概率 P（正确）对齐。

双任务范式（实验 2）。为了测试 GWT 预测，我们使用了带有双任务结构的 MiniGrid-Empty-5x5-v0 变体。每集（最多 32 步）包含：

1. 编码相位：代理观察到两个瞬态线索：“次要”线索（颜色）和“主线索”cue（对象类型）。
2. 延迟阶段：代理必须在工作记忆中保持这两个线索的同时，穿行空房间。
3. 报告阶段 1（早期）：特定的“报告标志”信号触发代理输出次级提示。
4. 报告阶段 2（延迟）：第二个标志触发代理输出主提示。

我们采用了“强病变”接线：提示信息仅通过工作区总线传输，而标准的循环编码器对这些特定提示通道是盲目的。这确保工作区是获取任务相关信息的唯一因果路径。

潜噪应力测试（实验 3）。为了评估双任务范式的鲁棒性（实验 2），我们引入了潜在工作空间扰动协议。我们没有在像素观测中添加噪声（传统卷积神经网络可以轻松过滤），而是在写操作后、读取操作之前，直接将高斯噪声 $N(0, \sigma)$ 注入代理内部工作区槽。对于无工作区的仅递发控制（仅 HOT），我们类似地在重复更新后、下游读出前，向递发载波状态（后 GRU 隐藏状态）注入高斯噪声。这种“脑刺激”方法测试系统对内部退化的稳定性。我们测量了 L75：报告步骤决策准确率降至 75%的噪声σ。这有效地量化了认知结构的“容忍余裕”。

4.3 培训

代理通过专家预言机的行为克隆（BC）进行训练，以确保基线能力。我们使用 BC 作为优化控制：它减少强化学习特有的混杂因素（探索失败、奖励稀疏和训练不稳定性），使病灶和扰动差异归因于结构操作而非优化动态。神谕仅用于训练期间生成行动目标;推理时从不参考它。所有评估均为所学策略的自主推广，我们的核心结果通过训练后因果干预（病灶、容量消融和潜在工作区噪声）应用于训练过的代理得出。这种框架使我们的主张带有条件：给定达到基线一阶能力的主体，哪些组成部分对访问和元认知相关签名是因果必不可少的?

[3] 我们使用“L75”而非“LD50”，以避免与毒理学惯例中 LD50 表示致死率 50%剂量的混淆。我们的阈值是 75%的准确率（不是 50%）。

4.3.1 Training Procedure

Dataset Generation. We generated training data using a hard-coded algorithmic solver (the "Oracle") that has perfect knowledge of the environment dynamics and task rules. For each experiment, we collected oracle demonstrations per agent type and seed (see Table 2 for episode counts by experiment). Each episode consists of up to 32 environment steps. The Oracle's action at each step was recorded as the target distribution for behavior cloning.

Loss Function. We minimized the Kullback-Leibler (KL) divergence between the agent's predicted action distribution $\pi_\theta(a|s)$ and the oracle's action distribution $\pi^*(a|s)$:

$$\mathcal{L}_{\text{BC}} = \mathbb{E}_{s\sim\mathcal{D}}\left[D_{\text{KL}}(\pi^*(a|s)\|\pi_\theta(a|s))\right]$$

where $\mathcal{D}$ is the dataset of oracle demonstrations.

Optimization. Training proceeded for the number of episodes specified in Table 2 (with one optimizer update per step) using the Adam optimizer [Kingma and Ba, 2014] with default momentum parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$). The learning rate was set to 10^{-3} for all agents. We used online training (processing each episode immediately rather than batching across episodes) to maintain temporal coherence of recurrent states. No learning rate scheduling or warm-up was employed.

Convergence Criteria. Training was run for a fixed number of episodes rather than early stopping, as behavior cloning from a deterministic oracle converges reliably. We verified convergence by confirming that agents achieved >95% oracle-action agreement on held-out episodes (per-step argmax match) by the end of training. Seeds that failed to reach this imitation threshold were flagged and excluded from analysis.

4.3.2 Stabilization Techniques

The dual-task paradigm creates two challenges for training: (1) temporal class imbalance, where report steps constitute only ∼3% of all steps, and (2) the need for the Self-Model to learn meaningful representations without direct supervision. We addressed these with targeted interventions:

- **Report-Step Reweighting:** Report steps are rare but critical for evaluation. To prevent these sparse signals from being dominated by navigation steps, we upweighted the KL loss on report steps by a factor of 3.0. Additionally, we applied class-specific weighting: the "primary cue" action (correct on only ∼5% of steps) received 20× higher weight than the "secondary cue" action in the loss computation.

- **Metacognitive Supervision (B2 agents; Experiments 1 and 3):** For B2 agents, we added a supervised auxiliary loss training the confidence head to predict binary correctness on report steps: $\mathcal{L}_{\text{meta}} = \text{BCE}(\hat{c}, \mathbf{1}_{\text{correct}})$, where $\hat{c}$ is the agent's confidence output. Here, $\mathbf{1}_{\text{correct}}$ is defined by the agent's own *intended* report decision (argmax of its policy logits) being correct, as assessed against the oracle-provided ground-truth label at that report step. This trains calibration on the agent's *own* failure modes ("will my imminent decision be correct?") rather than on predicting oracle behavior. This places the Self-Model (z_{self}) directly in the gradient path for metacognitive judgments.

- **Stimulus Auxiliary Loss (Experiment 1 only):** To ensure the workspace maintains stimulus information throughout the delay period, we added an auxiliary loss training a linear probe on the workspace state to predict stimulus presence at every step, not just report steps. This provides dense gradients that prevent the workspace memory trace from fading before the critical report moment.

4.3.3 Validation Strategy

We employed a simple held-out validation approach:

- **Train/Test Split:** After training on the specified number of oracle episodes (4,000 for E1/E2; 8,000 for E3; see Table 2), we evaluated agents on held-out independently sampled episodes with

4.3.1 训练程序

数据集生成。我们使用一个硬编码的算法求解器（“Oracle”）生成了训练数据，该算法对环境动态和任务规则有着完美的了解。对于每个实验，我们根据药物类型和种子收集了预言证（见表 2，按实验计量）。每集包含多达 32 个环境步骤。Oracle 在每一步的动作被记录为行为克隆的目标分布。

损失函数。我们最小化了智能体预测动作分布π（a|s）与预言机动作分布π（a|s）之间的 Kullback-Leibler（KL）散度：

$$L = E[D（\pi（a|s） \mathbin{/\!/} \pi（a|s））]$$

其中 D 是预言机演示数据集。

优化。训练按照表 2 中指定的发作数量进行（每步更新一次），使用 Adam 优化器[Kingma 和 Ba， 2014]，默认动量参数（β= 0.9，β= 0.999）。所有代理的学习率都设置为 10。我们使用在线训练（即时处理每个发作，而非跨发作批量处理）来保持反复发作状态的时间一致性。没有采用学习率的排班或热身。

收敛标准。训练以固定次数运行，而非提前停止，因为从确定性预言机进行行为克隆时会可靠收敛。我们通过确认代理在训练结束时对未完成的发作（每步 argmax 匹配）达成>95%的 oracle-action 一致性来验证收敛性。未达到这一模拟阈值的种子被标记并排除分析。

4.3.2 稳定技术

双任务范式为训练带来了两个挑战：（1）时间类不平衡，报告步骤仅占所有步骤的约 3%，（2）自我模型需要在没有直接监督的情况下学习有意义的表征。我们通过有针对性的干预措施应对了这些问题：

- 报告步骤重权：报告步骤虽罕见，但对评估至关重要。为防止这些稀疏信号被导航步骤主导，我们将报告步骤的 KL 损失加权 3.0 倍。此外，我们应用了类别特定的权重：“主提示”动作（仅在约 5%的步骤上正确）在损失计算中权重比“次要提示”动作高出 20×。

- 元认知监督（B2 代理;实验 1 和 3）：对于 B2 代理，我们增加了监督辅助损耗，训练置信头预测报告步骤的二元正确性：L= BCE（ ˆc， 1correct），其中 ˆc 是代理的置信输出。这里，1 正确定义为代理自身预期的报告决策（其策略 logit 的 argmax）正确，该决策基于该报告步骤 Oracle 提供的真实标签进行评估。这训练校准基于智能体自身的失效模式（“我即将做出的决策是否正确？”），而非预测预言机行为。这使得自我模型（z）直接处于元认知判断的梯度路径中。

- 刺激辅助丢失（仅实验 1）：为确保工作区在整个延迟期间保持刺激信息，我们增加了辅助损失，训练线性探针，预测每个步骤的刺激存在，而不仅仅是报告步骤。这提供了密集的梯度，防止工作区内存痕迹在临界报告时刻前消失。

4.3.3 验证策略

我们采用了一种简单的“保留验证”方法：

- 训练/测试分配：在对指定次数的预言机发作进行训练（E1/E2 为 4,000 发;E3 为 8,000 发;见表 2）后，我们对被保留的独立抽样发作进行了评估，受试者为

different random seeds (Experiment 2 capacity sweeps use 32 episodes per seed; robustness titrations use 50 episodes per noise level). These held-out episodes used the same task structure but different initial conditions and cue assignments.

- **Metrics:** Validation performance was assessed via oracle-action agreement (imitation accuracy) and task-level scores such as conjunction accuracy (both report phases correct) and report-window decision accuracy. We required >95% oracle-action agreement on held-out data to consider training successful.

- **No Hyperparameter Tuning on Test Data:** All hyperparameters were fixed before evaluation. The evaluation episodes used for reporting results (Experiments 1–3) were never used during hyperparameter selection.

4.3.4 Hyperparameter Selection

Hyperparameters were selected through a combination of theory-driven priors and limited manual search:

- **Architecture:** Workspace capacity ($K = 4$ slots) and dimension ($D = 16$) were chosen to match estimates of human visual working memory capacity [Luck and Vogel, 1997, Cowan, 2001]. The Self-Model latent dimension (64) was set to allow sufficient representational capacity while remaining no larger than the workspace state ($K \times D = 64$).

- **Learning Rate:** We tested $\{10^{-4}, 10^{-3}, 10^{-2}\}$ on a single seed and selected 10^{-3} based on fastest convergence without instability. This choice was fixed for all subsequent experiments.

- **Loss Weights:** The report-step reweighting factor (3.0) and class weight (20.0) were selected to approximately balance the contribution of rare and common step types to the total loss. We did not extensively tune these values.

- **No Automated Search:** We did not employ grid search, random search, or Bayesian optimization. The limited hyperparameter exploration reflects our goal of testing architectural hypotheses rather than maximizing benchmark performance.

Table 2 summarizes the key hyperparameters used across experiments.

Table 2: Training hyperparameters by experiment. All experiments used Adam optimizer with default momentum.

Parameter	**E1 (HOT)**	**E2 (GWT)**	**E3 (Triangulation)**
Training episodes	4,000	4,000	8,000
Steps per episode	32	32	32
Learning rate	10^{-3}	10^{-3}	10^{-3}
Workspace slots K	4	4 (varied)	4
Workspace dim D	16	16	16
Self-Model dim	64	—	64
Report step weight	3.0	3.0	3.0
Primary class weight	20.0	20.0	20.0
Meta-loss coefficient	1.0	—	1.0
Random seeds	20	20	20

4.4 Indicator Computation

Global Broadcast Index (GBI). We quantify how broadly information is shared across workspace slots using a participation-style coefficient. At each time step t, we treat each workspace slot as a node and define an adjacency matrix from slot similarity:

$$A_t(i, j) = \mathbf{w}_{t,i}^\top \mathbf{w}_{t,j}, \quad A_t(i, i) = 0$$

不同的随机种子（实验 2 容量扫描每个种子使用 32 集;鲁棒性滴定每噪声级别使用 50 次集数）。这些延期节目使用相同的任务结构，但初始条件和提示分配不同。

- 指标：通过 oracle-action 协议（模仿准确性）和任务级评分（如合取准确率（两阶段报告正确）和报告窗口决策准确性等评分来评估。我们要求对保留数据达>95%的 oracle-action 一致，才算是训练成功。

- 测试数据无超参数调优：所有超参数在评估前均已固定。用于报告结果的评估集数（实验 1–3）在超参数选择时从未使用。

4.3.4 超参数选择

我们通过理论先验和有限的人工搜索结合，评估了被保留的独立抽样发作，超参数为基础：

- 架构：工作空间容量（K = 4 个槽位）和尺寸（D = 16）被选定以匹配对人类视觉工作记忆容量的估计[Luck 和 Vogel，1997;Cowan，2001]。自建模潜维（64）被设定为允许足够的表示容量，同时不大于工作区状态（K × D = 64）。

- 学习率：我们在单个种子上测试了{10， 10， 10}，并根据收敛速度和不稳定性选出了 10 个。这一选择在后续所有实验中都被固定。

- 损失权重：报告步重加权因子（3.0）和类别权重（20.0）被选中，旨在大致平衡稀有步和常见步数对总损失的贡献。我们并没有对这些数值进行过深入调校。

- 无自动搜索：我们未使用网格搜索、随机搜索或贝叶斯优化。有限的超参数探索反映了我们测试架构假设而非最大化基准性能的目标。

表 2 总结了各实验中使用的关键超参数。

表 2：通过实验训练超参数。所有实验都使用带有默认动量的 Adam 优化器。

参数	E1（热点）	E2（大重测量）	E3（三角测量）
训练 每集 4,000	4,000	8,000 步	
	32	32	32
学习速率	10	10	10
Workspace slots K 4	4	（varied）	4 Workspace dim D
	16	16	16
自建模 dim	64	—	64
报告步进权重 3.0	3.0	3.0 主职业权重 20.0	20.0 元损失系数 1.0
—	1.0 随机种子 20	20	20

4.4 指示器计算

全球广播指数（GBI）。我们用参与系数量化信息在工作空间间的广泛共享程度。在每个时间步 t，我们将每个工作区槽视为节点，并根据槽相似度定义邻接矩阵：

$$A(i, j) = ww, \quad A(i, i) = 0$$

where $\mathbf{w}_{t,i}$ is the activation vector of slot i. For each slot i, we compute a participation coefficient

$$P_i(t) = 1 - \sum_{j \neq i} \left(\frac{A_t(i,j)}{\sum_{j' \neq i} A_t(i,j') + \varepsilon} \right)^2 .$$

GBI is the mean of $P_i(t)$ across slots and time within the decision window. Higher GBI indicates that each slot couples to multiple other slots (broad broadcast), while low GBI indicates narrow coupling/fragmentation.

Ignition Sharpness. We detected "ignition" events by looking for sudden jumps in the total magnitude of workspace activation. We computed the temporal derivative of the activation norm and measured the peak steepness. A sharp peak indicates an abrupt ignition-like transition; a flat trace indicates gradual or failed access.

Type-2 AUROC (Metacognition). We used the Area Under the Receiver Operating Characteristic curve for Type-2 judgments.

- **Input:** For every trial, we record the agent's binary correctness (0 or 1) and its continuous confidence wager $c \in [0, 1]$.
- **Analysis:** We treat confidence as a predictor of correctness. We sweep a threshold t from 0 to 1. For each t, we calculate the True Positive Rate (proportion of correct trials with $c > t$) and False Positive Rate (proportion of incorrect trials with $c > t$).
- **Result:** The area under this curve represents the probability that the agent assigns higher confidence to a correct trial than an incorrect one. 0.5 is chance (no insight); 1.0 is perfect insight.

Perturbational Complexity (PCI-A, ΔPCI). To probe an IIT-adjacent integration proxy, we used a perturbation approach inspired by the Perturbational Complexity Index (PCI).

1. **Perturb:** We inject a pulse of noise into the workspace.
2. **Measure:** We record the subsequent trajectory of workspace states.
3. **Compress:** We binarize the trajectory and compress it using the Lempel–Ziv algorithm (gzip-style compression).
4. **Calculate:** The file size of the compressed trajectory indicates its algorithmic complexity; we report this absolute score as raw PCI-A. Because absolute PCI-A can be inflated by unstructured failure/noise in engineered agents, we also report a within-agent contrast, $\Delta\text{PCI} = \text{PCI-A}_{\text{correct}} - \text{PCI-A}_{\text{incorrect}}$, which asks whether perturbational complexity is *selectively* higher on successful trials.

No-Report Signature (ΔNRS). To characterize whether stimulus information is present in the agent's internal state during the delay period, even when explicit report is blocked, we fit a linear decoder on internal feature vectors from blocked-report steps to predict the binary stimulus label. We score the decoder with AUC and subtract a shuffled-label baseline computed with the same cross-validation splits:

$$\Delta\text{NRS} = \text{AUC}_{\text{true}} - \mathbb{E}[\text{AUC}_{\text{shuffle}}].$$

A positive ΔNRS indicates above-baseline decodability; values near zero indicate little or no decodable signal, and negative values indicate below-baseline decoding due to finite-sample noise. In the bus-off condition, the workspace broadcast is identically zero, so ΔNRS reflects what can be decoded from the remaining trunk state alone.

其中 wis 是槽 i 的激活向量。对于每个槽 i，我们计算一个参与系数

$$P(t) = 1 - \sum_{j \neq i} P \frac{A(i, j)}{j \neq i \, A(i, j) + \varepsilon}!.$$

GBI 是决策窗口内 P（t）在时段和时间内的平均值。较高的 GBI 表示每个时隙与多个其他时隙耦合（宽广播），而较低的 GBI 表示狭窄耦合/分片。

点火锐利度。我们通过观察工作区激活总量的突然跳跃来检测“点燃”事件。我们计算了激活范数的时间导数并测量了峰值陡峭度。尖锐的峰值表示突如其来的点火式转变;平坦的走线表示访问过程渐进或失败。

2 型 AUROC（元认知）。我们使用了受试者工作特征曲线下的面积来进行 2 型判断。

- 输入：对于每个试验，我们记录代理人的二元正确性（0 或 1）及其连续置信度 c ∈ [0， 1]。

- 分析：我们将信心视为正确性的一个预测因子。我们将阈值 t 从 0 扫到 1。对于每个 t，我们计算真阳性率（正确试验中 c >t 的比例）和假阳性率（错误试验中 c >t 的比例）。

- 结果：该曲线下的面积代表代理人对正确试验赋予更高置信度的概率。0.5 是偶然（无洞察）;1.0 是完美的洞察。

微扰复杂性（PCI-A，ΔPCI）。为了探究 IIT 邻近的积分代理，我们采用了受扰动复杂性指数（PCI）启发的微扰方法。

1. 扰动：我们向工作空间注入一束噪声脉冲。
2. 测量：我们记录工作空间状态的后续轨迹。
3. 压缩：我们将轨迹二元化，并使用 Lempel–Ziv 算法（gzip 式压缩）进行压缩。
4. 计算：压缩轨迹的文件大小表明其算法复杂度;我们将该绝对分数报告为原始 PCI-A。由于工程代理中绝对 PCI-A 可能因非结构性失效/噪声而膨胀，我们还报告了代理内对比 ΔPCI = PCI-A− PCI-A 错误，该对比探讨成功试验中扰动复杂度是否选择性更高。

无报告签名（ΔNRS）。为了在延迟期间即使显式报告被阻断，也判断刺激信息是否存在于代理内部状态中，我们对阻断报告步骤的内部特征向量拟合线性解码器，以预测二元刺激标签。我们用 AUC 对解码器进行评分，并减去一个以相同交叉验证分割计算的洗牌标签基线：

$$\Delta \mathrm{NRS} = \mathrm{AUC} - E[\mathrm{AUC}]。$$

正值的 ΔNRS 表示可解码性高于基线;接近零的值表示可解码信号很少或没有，负值表示由于有限样本噪声导致的解码低于基线。在总线关闭状态下，工作区广播完全为零，因此 ΔNRS 反映的是仅从剩余中继状态中可以解码的内容。

4.5 Ablations and Manipulations

We employed precise causal interventions:

- **No-rewire Self-Model lesion (Experiment 1):** We evaluate the same trained checkpoint twice: intact and with the Self-Model removed (so z_{self} is replaced by zeros), while leaving the policy pathway unchanged. We use checkpoints whose policy reads the workspace directly (not exclusively z_{self}) so that first-order task performance remains intact under the lesion, allowing a clean Type-1/Type-2 dissociation test.
- **Capacity Ablation (Experiment 2):** We varied workspace capacity. For primary behavioral analyses we report three capacities (4, 2, or 0 slots; Figure 2). For the no-report signature (ΔNRS), we also ran a 5-level sweep of `workspace_capacity_scale` $\in \{0, 0.25, 0.5, 0.75, 1.0\}$ (0–4 effective slots) to test for a bus-on/bus-off discontinuity.

5 Results

5.1 Experiment 1 (HOT): Metacognitive Calibration Depends on the Self-Model

5.1.1 Motivation

Higher-Order Theories (HOT) of consciousness posit that a system becomes conscious of a state only when it represents itself as being in that state [Rosenthal, 2005, Lau and Rosenthal, 2011]. In artificial agents, this implies that effective metacognitive monitoring ("knowing that you know") requires a dedicated architectural component that re-represents the system's internal state. We selected this angle to test the **Monitoring** dimension of consciousness, specifically to dissociate first-order *doing* (task competence) from second-order *knowing* (metacognitive awareness), a distinction central to phenomena like blindsight.

In Experiment 1, we operationalize these predictions by equipping a global workspace agent with an explicit *Self-Model* (z_{self}) that compresses internal states into a latent summary. We test three key hypotheses:

1. **Metacognitive Necessity:** Accurate second-order confidence judgments (metacognition) require the Self-Model's re-representation of the workspace state.
2. **Causal Specificity:** Ablating the Self-Model should degrade metacognitive calibration (Type-2 AUROC) without impairing first-order task performance, demonstrating a double dissociation.
3. **Informational Content:** The Self-Model must actively compress and filter task-relevant information (e.g., uncertainty), rather than acting as a passive passthrough or random projection.

5.1.2 Results

Figure 1 presents the main findings. The Self-Model ablation produced a dissociation between first-order task performance and second-order metacognitive sensitivity.

First-Order Performance is Preserved. Task accuracy was largely unchanged between conditions. Across $n = 20$ random seeds, the self-on model achieved $88.5\% \pm 4.0\%$ accuracy (mean $\pm$ SD), while the self-removed model achieved $88.2\% \pm 3.7\%$ accuracy, a difference of only 0.3 percentage points (Figure 1a, left bars). This confirms that the no-rewire Self-Model lesion preserves first-order task performance while ablating metacognitive signal content.

Type-2 AUROC Collapses to Chance. Conversely, metacognitive sensitivity was reduced by Self-Model removal. The self-on condition achieved a mean Type-2 AUROC of 0.92 ($n = 20$ seeds; no-rewire evaluation), indicating well-above-chance metacognitive calibration. The self-removed condition sat exactly at chance (AUROC $= 0.50$ for all seeds), yielding a Self-Model effect of $\Delta\text{AUROC} = +0.42$ (95% CI $[0.38, 0.45]$). The ROC curves (Figure 1b) visualize this collapse: the self-on curve lies well above the diagonal, while the self-removed condition falls on the diagonal (AUROC $= 0.50$).

To put these numbers in context: an AUROC of 0.92 means the agent's confidence is higher on correct trials 92% of the time (vs. 50% by chance). The collapse to 0.50 under ablation indicates confidence

4.5 消融与操作

我们采用了精确的因果干预措施：

- 无重接自模型病变（实验 1）：我们对同一训练检查点进行两次评估：保持完整且移除自模型（即被零替换），但策略路径保持不变。我们使用策略直接读取工作区（而非仅 z）的检查点，以确保一阶任务性能在病变下保持完整，从而实现干净的 1 型/2 型解离检验。

- 容量消融（实验 2）：我们调整了工作空间容量。对于初级行为分析，我们报告三种能力（4、2 或 0 个槽位;图 2）。对于无报告签名（ΔNRS），我们还进行了 5 级扫描，分别为 workspace_capacity_scale ∈ {0， 0.25， 0.5， 0.75， 1.0}（0–4 个有效时隙），以检测母线开/断开。

5 选举结果

5.1 实验 1（HOT）：元认知校准依赖于自模型

5.1.1 动机

高阶意识理论（HOT）认为，只有当系统将自己表现为处于该状态时，系统才会意识到该状态[Rosenthal，2005;Lau 和 Rosenthal， 2011]。在人工代理中，这意味着有效的元认知监控（“知道你知道”）需要一个专门的架构组件来重新表示系统的内部状态。我们选择这一角度来检验意识的监测维度，特别是将一阶行为（任务能力）与二阶认知（元认知意识）分离，后者是盲视等现象的核心区别。

在实验 1 中，我们通过为全局工作空间代理配备显式的自模型（z），将内部状态压缩为潜在摘要，从而操作化这些预测。我们检验三个关键假设：

1. 元认知必然性：准确的二阶置信度判断（元认知）需要自我模型对工作区状态的再表征。

2. 因果特异性：自我模型的消融应能降低元认知校准（2 型 AUROC），同时不影响一阶任务表现，显示双重解离。

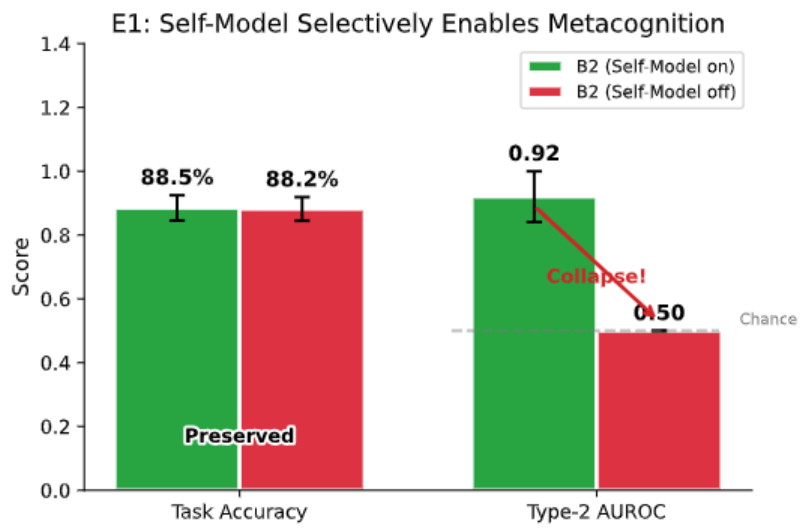
3. 信息内容：自模型必须主动压缩和过滤与任务相关的信息（例如不确定性），而不是作为被动的穿透或随机投射。

5.1.2 选举结果

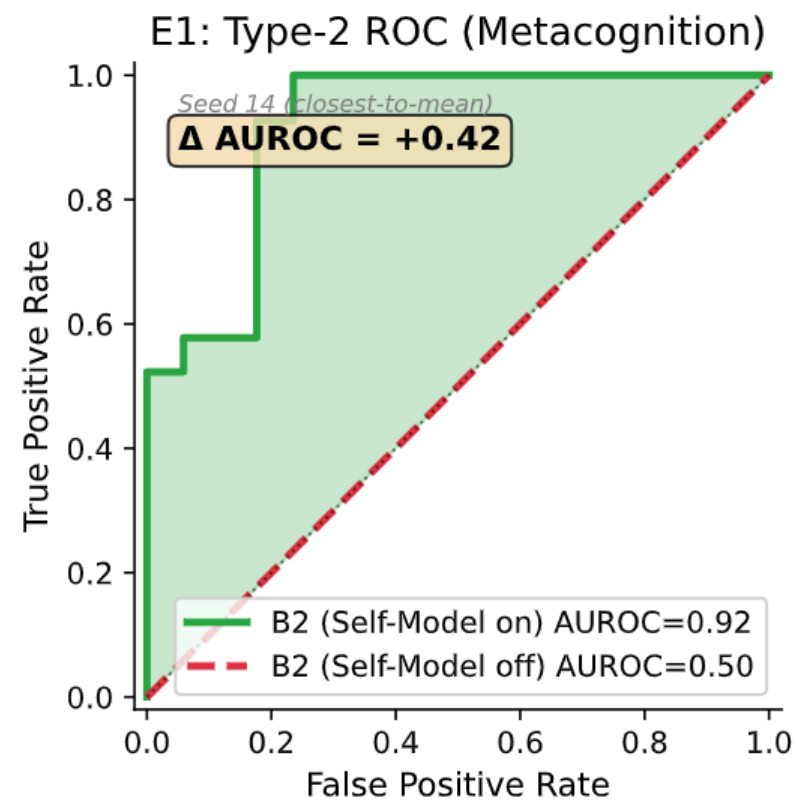
图 1 展示了主要发现。自模型消融导致一阶任务表现与二阶元认知敏感性之间产生了分离。

一级性能得以保留。任务精度在不同条件下基本没有变化。在 n=20 个随机种子中，自取模型实现了 88.5%±4.0%的准确率（平均标准差±），而自移除模型实现了 88.2%±3.7%的准确率，仅相差 0.3 个百分点（图 1a，左边条）。这证实了无重接自模型病变在消解元认知信号内容的同时，保持了一阶任务表现。
二型 AUROC 崩溃成偶然。相反，元认知敏感性通过自我模型去除降低。自我启动条件的平均 2 型 AUROC 为 0.92（n=20 粒种子;无重接评估），表明元认知校准远超偶然。自移除条件恰好处于偶然状态（所有种子的 AUROC = 0.50），产生自模型效应ΔAUROC = +0.42（95% CI [0.38， 0.45]）。ROC 曲线（图 1b）直观地展示了这种塌陷：自上曲线远高于对角线，而自移除条件位于对角线上（AUROC = 0.50）。

为了让这些数字有个背景：AUROC 为 0.92 意味着代理人在正确试验中置信度更高（而偶然率为 50%）。消融后降至 0.50 显示信心



(a)



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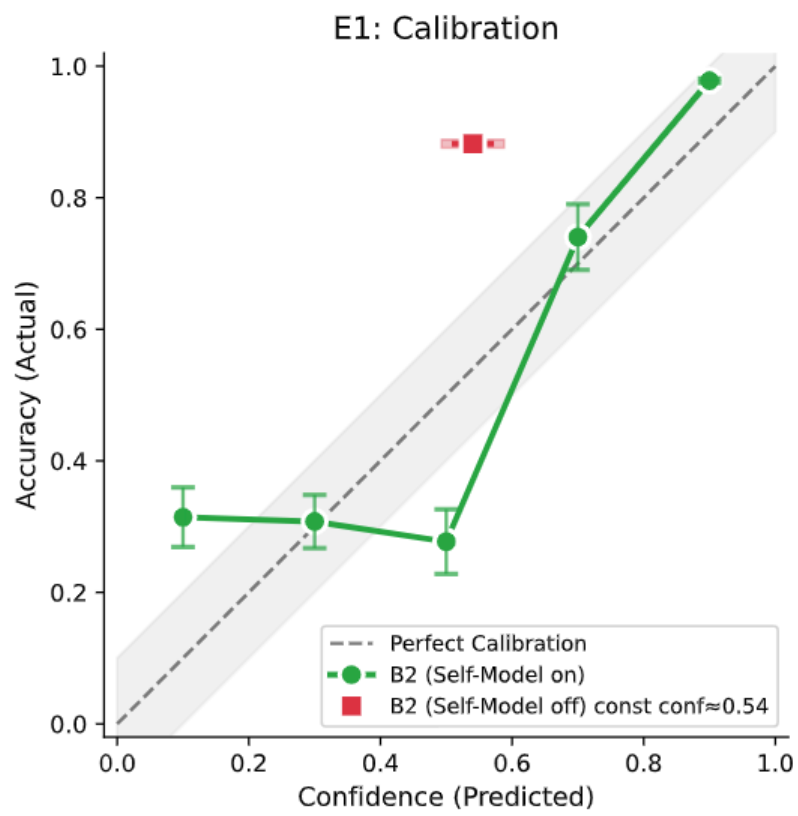

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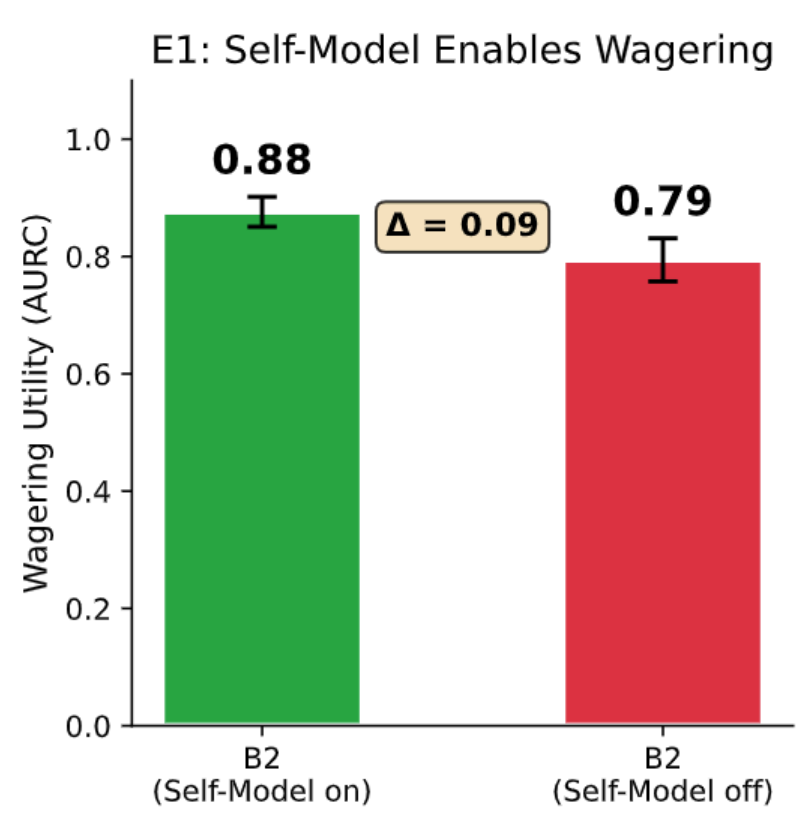
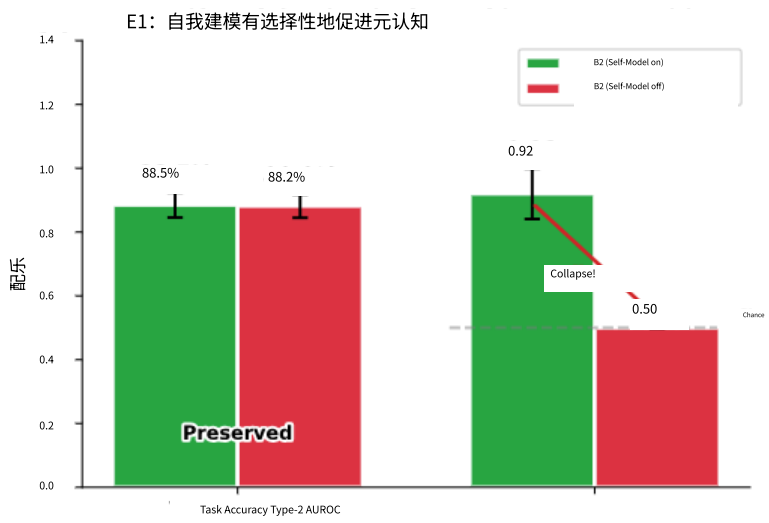


(d)

Figure 1: **Metacognitive calibration depends on the Self-Model.** (a) First-order task accuracy (left) is preserved under Self-Model ablation, while Type-2 AUROC (right) collapses to chance. Error bars show ±1 SD across 20 seeds. (b) Type-2 ROC curves: self-on (green) achieves AUROC = 0.92; self-removed (red) falls on the diagonal (AUROC = 0.50). Shaded region shows metacognitive advantage. (c) Calibration plot: self-on shows a positive confidence–accuracy relationship; self-removed outputs constant confidence (≈ 0.54). (d) Wagering utility (AURC): self-on achieves higher selective accuracy through calibrated confidence.



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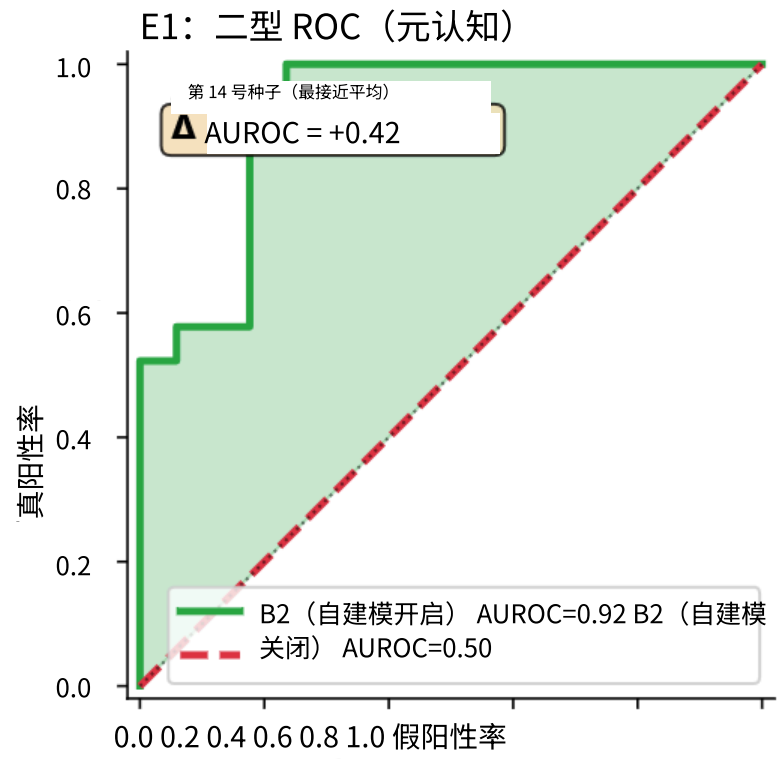

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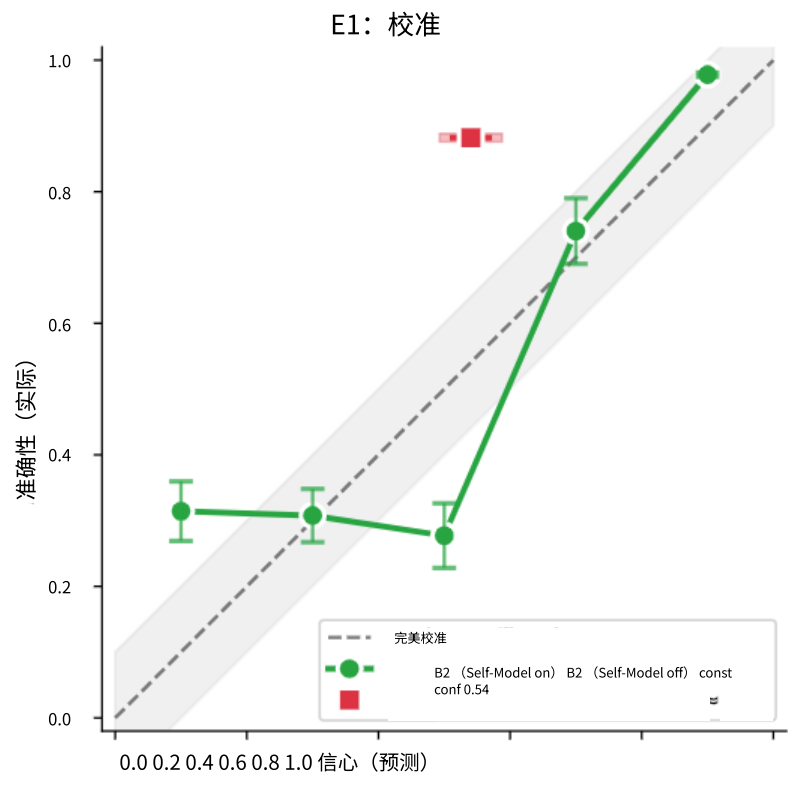

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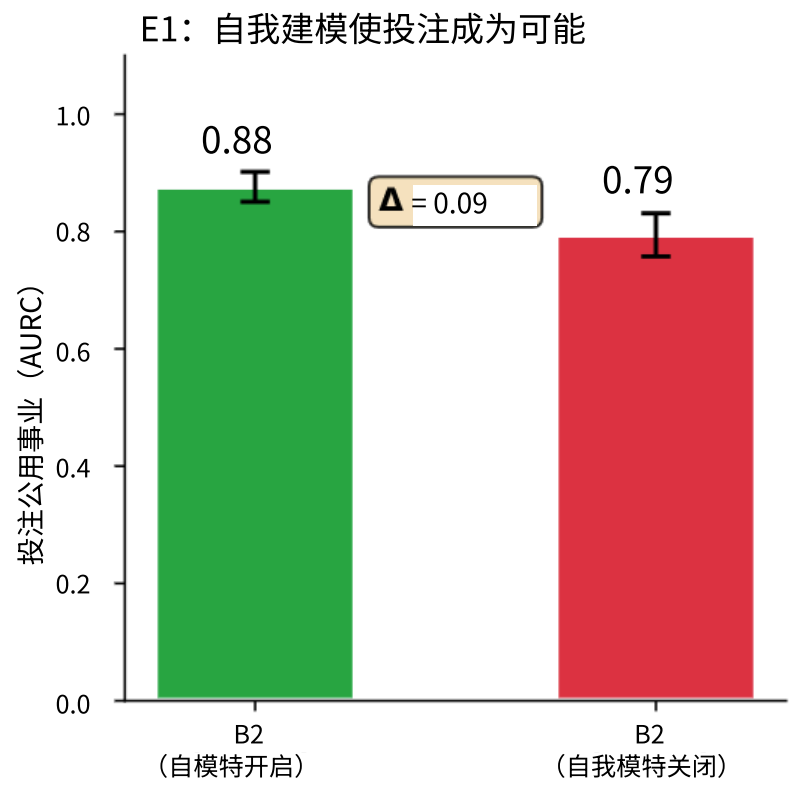

(d)

图 1：元认知校准依赖于自我模型。（a） 一阶任务精度
消融下的 50 表示置信度（左）在自模型消融下保持，而 2 型 AUROC（右）则坍缩为偶然。错误条显示 20 个种子中有±1 个 SD。（b） 2 型 ROC 曲线：自合（绿色）获得 AUROC = 0.92;自移除（红色）落在对角线上（AUROC = 0.50）。阴影区域显示了元认知优势。（c） 校准图：自开显示正置信-准确性关系;自移除输出恒定置信度（≈ 0.54）。（d） 投注效用（AURC）：自投通过校准置信实现更高的选择准确率。

ratings became uninformative about performance. This pattern mirrors human blindsight, where patients perform visual discriminations above chance while lacking subjective awareness, and thus any basis for confidence judgments.

Confidence Becomes Constant Without Self-Model. Examination of confidence distributions elucidated the mechanism underlying the AUROC collapse: in the self-removed condition, the confidence head outputs a near-constant value (0.54 ± 0.02). Without input from z_{self}, the confidence head has no information to modulate its output (Figure 1c).

Workspace Lesion Produces Chance-Level Confidence. A distinct question is how the intact Self-Model behaves when the global workspace itself is lesioned ($K = 0$; bus-off). One might worry that removing the monitored substrate could yield spuriously high confidence ("nothing contradicts me"), rather than low/flat confidence ("nothing to monitor"). We tested this directly by evaluating B2 with the Self-Model enabled under a bus-off lesion (`workspace_capacity_scale` $= 0.0$). Metacognitive sensitivity collapsed to near chance: Type-2 AUROC $= 0.492 \pm 0.058$ across $n = 20$ seeds (mean $\pm$ SD), with a higher opt-out rate (skip fraction $= 0.136 \pm 0.128$). By contrast, at full workspace capacity ($K = 4$; bus scale 1.0), metacognitive calibration is high (Type-2 AUROC $= 0.92$ in the main no-rewire evaluation; Figure 1b). Under the seed-level aggregation used for this workspace-lesion control (imputing AUROC $= 0.50$ when AUROC is undefined due to perfect first-order performance in a few seeds), this baseline is Type-2 AUROC $= 0.856 \pm 0.180$ with skip fraction $= 0.044 \pm 0.052$. Thus, lesioning the workspace does not produce "uninhibited" high confidence; it produces an effectively uninformative confidence signal consistent with the absence of a monitorable global state.

Control Analyses: Blind Ablation and Compression. Blind ablations (noise/permutation) also resulted in AUROC collapsing to ≈ 0.50, confirming that the specific *information content* of z_{self} is necessary, not merely the presence of an active pathway. Furthermore, analysis of z_{self} revealed an effective dimensionality of ≈ 2.0 (compressed from 89 input dimensions: 64+16+7+2), representing a $\sim$98% reduction in dimensionality. This compression is functionally significant: the Self-Model distills the high-dimensional workspace state into just two independent axes of variation, yet these two dimensions carry sufficient information to support metacognitive judgments. Despite this $\sim$98% compression, z_{self} outperformed a PCA baseline in predicting correctness (+5.1% AUROC advantage), indicating that the Self-Model performs task-relevant filtering rather than merely preserving the directions of maximum variance. In other words, the Self-Model learns *what matters* for metacognition, not just what varies most.

Deep Dive: Workspace Collapse and Passthrough. We conducted a deep dive into failure modes. In one outlier seed where the self-on agent failed to achieve calibration (AUROC ≈ 0.5), analysis revealed that the underlying workspace slots contained no information about correctness (Cohen's $d \approx 0$). A linear probe on the workspace bus outperformed the Self-Model in this case, suggesting the Self-Model acts as a non-linear filter: when the workspace signal is strong, z_{self} amplifies it for metacognition; when the signal is weak, z_{self} collapses, effectively destroying the information. This reinforces the hierarchical dependency: the Self-Model cannot read what the workspace has not written.

We next ask whether the underlying workspace capacity itself determines information access, independent of higher-order monitoring.

5.2 Experiment 2 (GWT): Workspace Capacity Determines Information Access

5.2.1 Motivation

Global Workspace Theory (GWT) posits that conscious access corresponds to a non-linear "ignition" event in which information becomes globally available to distributed modular systems via a capacity-limited central resource [Baars, 1988, Dehaene and Changeux, 2011]. While neural correlates of ignition are well documented in biological brains [Mashour et al., 2020], causal evidence linking workspace capacity, ignition dynamics, and behavioral information access remains sparse. We selected this angle to test the **Access** dimension of consciousness: determining whether the workspace is merely a passive correlate or the *causal substrate* of information integration and broadcast.

In Experiment 2, we operationalize GWT's core predictions in an artificial agent to test whether:

收视率对表现变得缺乏信息。这种模式类似于人类盲视，患者在缺乏主观意识的情况下，凭空进行视觉辨别，因此缺乏信心判断的依据。
没有自我榜样，自信会变得恒常。对置信分布的分析阐明了 AUROC 坍缩的机制：在自移除条件下，置信头输出的值接近恒定（0.54±0.02）。没有来自 z 的输入，置信头没有信息来调制其输出（见图 1c）。

工作区病变产生机会级信心。一个独特的问题是，当全局工作空间本身被损伤（K = 0;总线关闭）时，完整的自模型会如何表现。有人可能会担心移除监测基底会得到虚假的高置信度（“没有矛盾”），而不是低置信度（“无可监测”）。我们通过在公交线断开病灶（workspace_capacity_scale = 0.0）下，直接评估启用自模型的 B2 来测试。元认知敏感性几乎是偶然：2 型 AUROC=0.492±0.058，n=20 个种子（平均标准差±），选择退出率更高（跳过分数 =0.136±0.128）。相比之下，在满工作空间容量（K = 4;总线尺度 1.0）时，元认知校准较高（2 型 AUROC = 0.92，主要无重线评估;图 1b）。在用于该工作区-病灶控制的种子层聚合（当 AUROC 因少数种子一阶表现完美而未定义时，估算 AUROC = 0.50），该基线为 2 型 AUROC = 0.856 ± 0.180，跳跃分数=0.044 ± 0.052。因此，对工作区进行病变并不会产生“无抑制”的高置信度;它产生了一个实际上无信息的置信信号，与缺乏可监测的全局状态相符。

对照分析：盲眼消融与压迫。盲消融（噪声/置换）也导致 AUROC 坍缩至≈0.50，证实了所需的特定信息量，而不仅仅是活性通路的存在。此外，z 的分析显示有效维度为≈ 2.0（从 89 个输入维度压缩而成：64+16+7+2），代表维度降低了约 98%。这种压缩在功能上具有重要意义：自模型将高维工作空间状态提炼为仅两个独立的变异轴，但这两个维度携带的信息足以支持元认知判断。尽管压缩率约为 98%，z 在预测正确性方面表现优于 PCA 基线（+5.1% 的 AUROC 优势），表明自模型执行的是任务相关过滤，而不仅仅是保持最大方差方向。换句话说，自我模型学习的是对元认知重要的内容，而不仅仅是变化最大的部分。

深度潜探：工作区崩溃与直通。我们深入探讨了故障模式。
在一个异常种子中，自启动代理未能实现校准（AUROC ≈ 0.5），分析显示底层工作区槽不包含正确性信息（Cohen's d ≈ 0）。在这种情况下，工作区总线上的线性探针表现优于自模型，表明自模型起到非线性滤波器的作用：当工作区信号强时，将其增强以进行元认知;当信号微弱时，Z 崩溃，有效地销毁信息。这强化了层级依赖关系：自模型无法读取工作区未写入的内容。
接下来我们要问的是，底层的工作空间容量本身是否决定了信息的获取，而这与高阶监控无关。

5.2 实验 2（GWT）：工作区容量决定信息访问

5.2.1 动机

全球工作空间理论（GWT）认为，有意识访问对应于一个非线性的“点火”事件，即信息通过容量有限的中央资源向分布式模块系统全局开放[Baars， 1988;Dehaene and Changeux， 2011]。虽然点火的神经相关性在生物大脑中有充分记录[Mashour 等，2020]，但将工作空间容量、点火动力学和行为信息获取联系起来的因果证据仍然稀少。我们选择这一角度来测试意识的访问维度：确定工作空间是仅仅被动的相关物，还是信息整合和广播的因果基底。

在实验 2 中，我们将 GWT 的核心预测应用于人工代理，以测试：

1. **Causal Necessity:** We test whether access to information for multi-step tasks depends causally on workspace capacity.
2. **Ignition Dynamics:** Successful access is marked by a sudden, non-linear increase in workspace activation ("ignition") and global sharing (broadcast).
3. **Ablation Sensitivity:** Reducing workspace capacity degrades both the neural signature (ignition/broadcast) and the behavioral outcome (task accuracy) in tandem.

5.2.2 Results

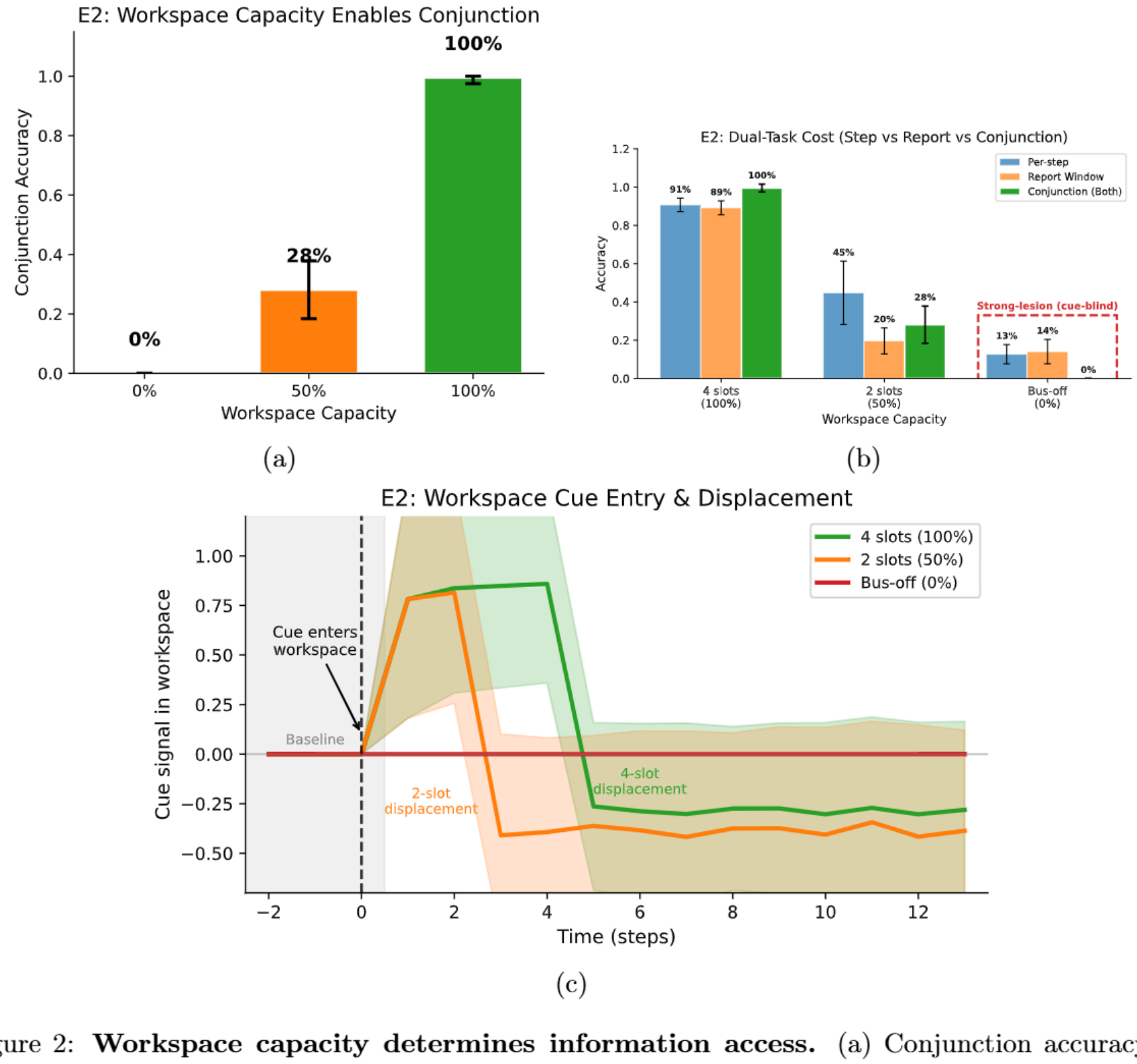

Figure 2: **Workspace capacity determines information access.** (a) Conjunction accuracy by workspace capacity ($n = 20$ seeds; mean $\pm$ SD): $99.5\% \pm 2.0\%$ at 4 slots, $28.1\% \pm 9.7\%$ at 2 slots, $0.0\% \pm 0.0\%$ at bus-off. Full vs. half capacity: Welch's t-test $p < 10^{-16}$ (seed-level), Hedges' $g = 9.70$. Baseline agents (A0, A1) achieve near-chance under strong-lesion wiring (not shown; see text); B1 at full capacity strongly outperforms both baselines ($g > 3$). (b) Dual-task cost summary: per-step accuracy, report-window decision accuracy, and conjunction success by capacity. Conjunction collapses at bus-off; the report-window decision is most sensitive to capacity reduction. (c) Workspace cue entry and displacement under backward masking. Cue signal (y-axis) jumps at cue entry, persists for a capacity-dependent duration, then decays as distractors displace cue content. Four slots: signal persists through 4 masks; two slots: faster displacement; bus-off: no signal (flat at 0). Note: A finer 5-level capacity sweep (0%, 25%, 50%, 75%, 100%) was conducted for the ΔNRS analysis (see text) but is not shown here.

Figure 2 presents the main findings across neural and behavioral measures.

Behavioral Performance Scales with Capacity. Conjunction accuracy showed a monotonic relationship with workspace capacity (Figure 2a). Across $n = 20$ random seeds (32 evaluation episodes per

1. 因果必然性：我们测试多步骤任务中信息获取是否因果依赖于工作空间容量。

2. 点燃动态：成功的访问标志着工作区激活（“点火”）和全局共享（广播）的突然非线性增加。

3. 消融敏感性：减少工作空间容量会同时降低神经特征（点燃/广播）和行为结果（任务准确性）。

5.2.2 选举结果

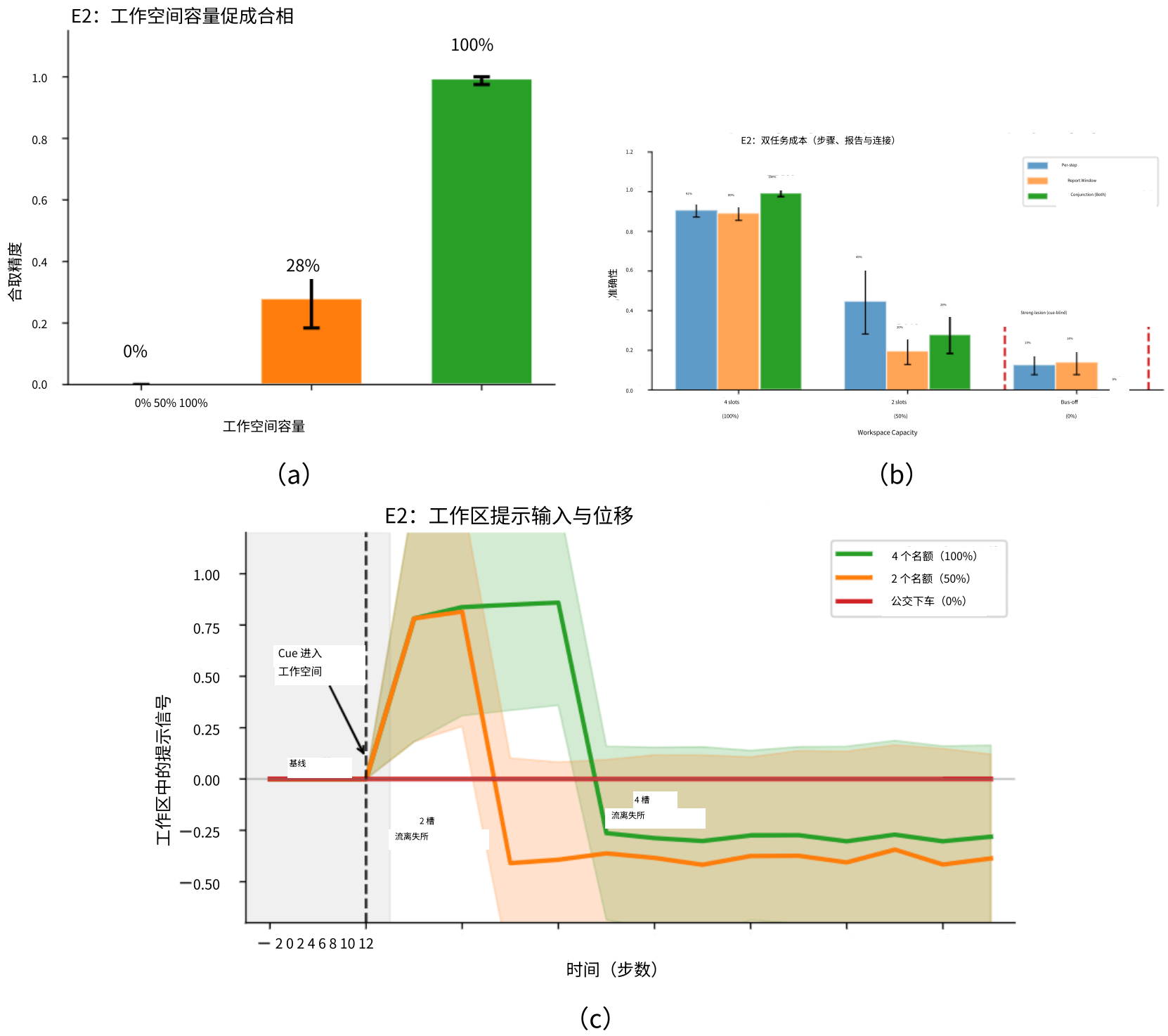

图 2：工作空间容量决定信息访问。（a） 合取精度
工作区容量（n = 20 个种子;平均 ± SD）：4 个槽位时 99.5% ±2.0%，2 个槽位±28.1% 9.7%，公交车关闭时 0.0% ±0.0%。满负荷与半容量：Welch's t 检验 p<为 10（种子级），Hedges 的 g=9.70。基线药物（A0， A1）在强病变线路下实现近似概率（未显示;见正文）;B1 在满负荷时表现明显优于两种基线（g>3）。（b） 双任务成本总结：每步准确率、报告窗口决策精度及按容量计算的连通成功率。合体在公交车下车处坍缩;报告窗口决策对容量减少最为敏感。（c） 工作区提示进入和后退掩蔽下的位移。提示信号（y 轴）在提示进入时跳跃，持续一段时间取决于容量，然后随着干扰物位移提示内容而衰减。四个槽口：信号通过 4 个掩码持续存在;两个槽位：更快的排水量;公交车关闭：无信号（0 点平稳）。注：通过更细致的五级容量扫描，我们将 GWT 的核心预测操作化为人工代理，以检验：（0%、25%、50%、75%、100%）是否进行了ΔNRS 分析（见正文），但此处未显示。

图 2 展示了神经和行为测量的主要发现。

行为表现随能力而提升。合取准确度显示与工作空间容量呈单调关系（图 2a）。跨越 n=20 个随机种子（每个 32 个评估集）

seed), at full capacity (4 slots), B1 achieved $99.5\% \pm 2.0\%$ conjunction accuracy (mean $\pm$ SD across seeds). At reduced capacity (2 slots), performance dropped to $28.1\% \pm 9.7\%$, showing substantial seed-to-seed variability but a clear degradation. At bus-off (0 slots), conjunction accuracy collapsed to $0.0\% \pm 0.0\%$; the task became unsolvable without workspace access to cue information.

Statistical validation: The unit of analysis for all statistical tests is the random seed ($n = 20$ per condition), treating each independently trained agent as a single observation (each seed contributes its mean over 32 evaluation episodes). We acknowledge that even this sample size limits power for moderate effects and yields nontrivial uncertainty around estimated thresholds. Concretely, with $n = 20$ per group, a two-sample comparison is still underpowered for modest effects: a standardized effect on the order of Cohen's $d \approx 0.9$ is typically required to reach 80% power at $\alpha = 0.05$ under roughly Gaussian seed-to-seed variability. Accordingly, we interpret the strongest evidence as qualitative, seed-consistent regime changes (e.g., near-ceiling vs. collapsed-to-zero performance) rather than precise estimates of thresholds or effect sizes; we report uncertainty for transparency and treat effect sizes as descriptive. For transparency, we also report p-values in coarse scientific notation (one significant digit) and interpret them as descriptive rather than as precise evidence of a critical threshold. The difference between full capacity and half capacity was statistically significant (Welch's t-test: $t = 31.3$, $p < 10^{-16}$; Hedges' $g = 9.70$, large effect). A non-parametric permutation test confirmed the result ($p < 10^{-4}$). Comparison against baseline agents was also decisive: B1 at full capacity (per-step accuracy $90.7\% \pm 3.5\%$) outperformed A0 (16.1%; $p < 10^{-9}$, $g = 3.30$) and A1 (15.2%; $p < 10^{-9}$, $g = 3.54$). The similar near-chance performance of A0 and A1 is expected under strong-lesion wiring: because cues are routed exclusively through the workspace bus (which neither baseline has), both agents are effectively guessing regardless of whether they have recurrent memory.

The 4-slot workspace is analogous to classic findings on human visual working memory capacity [Luck and Vogel, 1997, Cowan, 2001], which suggest a limit of approximately 3–4 items. Our results demonstrate that reducing this capacity has behavioral consequences: with only 2 slots available, agents frequently failed to maintain both cues across the delay period, mirroring the interference effects observed when humans are asked to hold more items than their capacity allows.

Bus-Off Discontinuity in an Access Marker. Beyond the linear scaling of performance, we analyzed the no-report signature (ΔNRS; Methods) across a 5-level capacity sweep (`workspace_capacity_scale` $\in \{0, 0.25, 0.5, 0.75, 1.0\}$; $n = 5$ seeds). Across the tested settings, all tested non-zero capacities (25–100%) produced strong above-baseline decodability ($\Delta\text{NRS} \approx 0.51$), while the complete workspace lesion (bus-off; 0%) collapsed to near-zero / below-baseline decoding ($\Delta\text{NRS} \approx -0.12$). Given the coarse, discrete nature of this sweep, we interpret this as a qualitative discontinuity at a complete lesion (bus-on vs. bus-off), not as evidence for a precisely estimated threshold or for continuous thresholding dynamics. We do not claim critical scaling, a thermodynamic-limit phase transition, or a precisely estimated critical point; rather, the pattern indicates that in this architecture and task, access-related dynamics are robust to partial capacity reductions but fail catastrophically when the workspace is removed.

Dual-Task Cost Reveals Decision-Specific Demands. Figure 2b summarizes performance at three granularities: per-step accuracy, report-window decision accuracy, and conjunction success. At full capacity, B1 achieves high per-step accuracy ($90.7\% \pm 3.5\%$) and report-window decision accuracy ($89.2\% \pm 3.6\%$), yielding near-ceiling conjunction accuracy ($99.5\% \pm 2.0\%$). At half capacity, per-step accuracy degrades ($44.8\% \pm 16.6\%$) and report-window decision accuracy is low ($19.6\% \pm 6.8\%$), producing only $28.1\% \pm 9.7\%$ conjunction accuracy. At bus-off, per-step accuracy is low ($12.7\% \pm 5.0\%$), conjunction accuracy collapses to $0.0\% \pm 0.0\%$, and report-window decision accuracy is low ($14.1\% \pm 6.4\%$), consistent with the absence of a functional access pathway for cue information. For interpretation, report-window decision accuracy has a $1/7 \approx 14.3\%$ chance floor (7 discrete actions), whereas conjunction accuracy requires both reports to be correct (chance $\approx (1/7)^2 \approx 2.0\%$ under independent guessing); thus, "0.0%" indicates that no successful conjunction episodes were observed rather than implying a 0% per-decision chance rate.

Cue Signal Dynamics Under Backward Masking. To visualize workspace dynamics at finer temporal resolution, we employed a backward masking paradigm (Figure 2c), a technique widely used in consciousness research to probe the temporal dynamics of conscious access [Dehaene, 2014]. In visual backward masking, a brief target stimulus is followed by a mask that interferes with its conscious perception; here, we present distracting stimuli after the cue to test how long the cue representation persists in the workspace before being displaced.

种子），满载（4 个槽位）时，B1 实现了 99.5%±2.0%的合相准确率（跨种子的平均标准偏差±）。在减少产能（2 个槽位）时，表现降至 28.1%±9.7%，显示出种子间差异较大，但明显退化。在总线关闭（0 槽位）时，连接精度降至 0.0%±0.0%;没有工作区获取提示信息，任务变得无法解决。

统计验证：所有统计检验的分析单位是随机种子（n=20 个，每个条件），将每个独立训练的代理视为单一观察值（每个种子平均贡献 32 次评估集数）。我们承认，即使样本量如此，也会限制中度效应的功效，并且在估计阈值附近产生非平凡的不确定性。具体来说，每组 n=20，二次样本比较在适度效果下仍然显弱：在种子间变异性大致上，通常需要以 Cohen's d≈0.9 为单位的标准效应才能达到 80%的幂α = 0.05。因此，我们将最强证据解释为在 n=20 个随机种子间的定性、种子一致的制度变化（每场 32 次评估（例如，接近上限与崩溃至零的表现），而非阈值或效应量的精确估计;我们报告透明度的不确定性，并将效应大小视为描述性。为了透明起见，我们还以粗略科学记号法（一个有效位）报告 p 值，并将其解释为描述性，而非关键阈值的精确证据。满载能力与半容量之间的差异具有统计学意义（韦尔奇 t 检验：t = 31.3，p < 10;赫奇斯 g = 9.70，影响较大）。非参数置换检验证实了结果（p < 10）。与基线药物的比较也具有决定性意义：满负荷 B1（每步准确率 90.7%±3.5%）优于 A0（16.1%;p < 10，g = 3.30）和 A1（15.2%;p < 10，g = 3.54）。 在强病变接线下，A0 和 A1 的近似偶然表现是预期的：由于线索仅通过工作区总线路由（两个基线均无此类总线），无论是否有重复记忆，两者实际上都是在猜测。

4 格工作区类似于关于人类视觉工作记忆容量的经典发现[Luck 和 Vogel， 1997，Cowan， 2001]，建议限制在大约 3–4 个项目之间。我们的结果表明，降低这种容量会带来行为后果：由于只有两个槽位，代理经常无法在整个延迟期间同时保持这两个线索，这与当人类被要求持有超过其容量允许的物品时观察到的干扰效应相呼应。

出入口标志的公交车出口不连续。除了性能的线性扩展，我们还分析了无报告签名（ΔNRS;方法）在 5 级容量扫描（workspace_capacity_scale ∈ {0， 0.25， 0.5， 0.75， 1.0};n = 5 个种子）中。在测试的设置中，所有测试的非零容量（25–100%）均表现出强的基础以上解码率（ΔNRS ≈ 0.51），而完整的工作区病变（总线关闭;0%）则坍缩至接近零/低于基线的解码率（ΔNRS ≈ −0.12）。鉴于该扫描的粗略离散性质，我们将此解释为完全病灶处的定性不连续（母线开通与母线关闭），而非精确估计阈值或连续阈值动态的证据。我们不主张临界缩放、热力学极限相变或精确估计的临界点;相反，该模式表明，在该架构和任务中，访问相关的动态对部分容量减少具有鲁棒性，但当工作区被移除时会灾难性失效。

双任务成本揭示了决策特定的需求。图 2b 总结了三个细度的表现：每步准确率、报告窗口决策精度和连通成功率。在满负荷状态下，B1 实现了高每步准确率（90.7%±3.5%）和报告窗口决策准确率（89.2%±3.6%），实现接近天花板的合相准确率（99.5%±2.0%）。在半容量时，每步准确率下降（44.8%±16.6%），报告窗口决策准确率较低（19.6%±6.8%），仅产生 28.1%±9.7%的连词准确率。在总线关闭时，每步准确率较低（12.7%±5.0%），连接准确率降至 0.0%±0.0%，报告窗口决策准确率较低（14.1%±6.4%），这与缺乏功能性线索信息访问路径相符。从解释角度看，报告窗口决策准确率为 1/7≈14.3%的概率底线（7 个离散动作），而连词准确率要求两个报告都正确（概率≈（1/7）≈独立猜测下为 2.0%）;因此，“0.0%”表示未观察到成功的合相事件，而非意味着每次决策的概率为 0%。

这就说明了信号动态在反向掩蔽下的表现。为了以更细致的时间分辨率可视化工作空间动态，我们采用了反向掩蔽范式（见图 2c），这是一种在意识研究中广泛使用的技术，用以探究意识访问的时间动态[Dehaene， 2014]。在视觉反向掩蔽中，短暂的目标刺激之后会有一个干扰其意识感知的掩膜;在这里，我们在提示后呈现干扰刺激，以测试提示表示在工作区中持续多久才被移位。

We utilized a "Report-Flag Signal" metric, exploiting the finding that the `report_flag` component creates strong separation in slot space (cosine similarity ~ 0.22 vs. ~ 0.98). This allowed us to construct a cue-specific projection that is positive when the cue is present and negative when displaced.

The results revealed capacity-dependent persistence of the cue signal:

- **4 slots (100%):** Signal jumped from 0 to $\sim$0.78 at cue entry, remained positive through 4 mask steps, then decayed to negative values (~ -0.26) as slot 0 was overwritten.
- **2 slots (50%):** Identical initial signal ($\sim$0.78), but faster decay; signal became negative by mask step 2 (~ -0.41).
- **Bus-off (0%):** Signal remained at 0 throughout (no workspace storage).

This pattern is analogous to human backward masking experiments: in both cases, the initial encoding of the target is identical across conditions (the $\sim$0.78 signal at cue entry), but the *duration* of conscious access differs based on capacity constraints. Just as a stronger mask or shorter stimulus-onset asynchrony (SOA) reduces conscious report in humans, reduced workspace capacity shortens the window during which information remains accessible for report in our agents. This demonstrates the GWT prediction: workspace capacity determines how long task-relevant information persists before displacement by incoming stimuli.

Neural Signatures Track Capacity. GBI at full capacity averaged 0.66, indicating broad cross-slot coupling (high participation across slots) consistent with the "global broadcast" intuition in GWT. At half capacity, GBI dropped to 0.03 (reduced broadcast breadth; lower cross-slot participation), reflecting the diminished opportunity for information to be shared across the bus with fewer slots available. Ignition sharpness showed the expected pattern: IS ≈ 0.15 at full capacity (bootstrap 95% CI [0.06, 0.26] across $n = 20$ seeds), IS ≈ 0.06 at half capacity (bootstrap 95% CI [0.02, 0.10]), and IS $= 0$ at bus-off (no ignition without a workspace).[4] The complete absence of ignition at bus-off demonstrates that ignition is not merely correlated with workspace activity but *depends* on it. Without the workspace, there is no substrate for the non-linear amplification that characterizes conscious access in GWT.

Bus Audit Confirms Causal Dependence. The "bus audit" (see Methods) confirmed strong gradients through the bus pathway (T1 median Jacobian norms $\sim$ 78–175) and zero leakage through the trunk (T3 trunk isolation 0% flips), validating that behavior genuinely depended on workspace contents.

Given that metacognitive access depends on a Self-Model and information access depends on capacity, we now ask whether these indicators predict functional robustness to internal noise.

5.3 Experiment 3 (Triangulation): Hierarchical Robustness and Composite Markers

5.3.1 Motivation

Theories of consciousness are often treated as competitors, each proposing a unique neural signature. However, our previous experiments suggest they describe distinct functional layers. We selected this final angle to test the **Integration** and **Functional Utility** dimensions: linking abstract neural markers (ignition, complexity, calibration) to the concrete evolutionary advantage of robustness. By bridging this "explanatory gap," we aim to show what consciousness is *for*.

In Experiment 3, we test a *Triangulation Hypothesis*: that a reduced composite of cross-architecture markers (GBI + ΔPCI) predicts robustness better than either term alone, and that adding a HOT-aligned metacognition term (Type-2 AUROC) provides incremental information *within* Self-Model agents. We explicitly evaluate whether these markers capture redundant information or contribute unique variance to robustness.

5.3.2 Results

Experiment 3 yields two co-equal findings: (1) a negative result constraining the transfer of PCI-style complexity proxies to engineered agents, and (2) a robustness hierarchy distinguishing broadcast-only architectures from those with higher-order monitoring. We present the PCI transfer constraint first because it carries direct implications for IIT-adjacent assessment in artificial systems.

[4] We note that bootstrap confidence intervals can become narrow when seed-to-seed variance is low. These narrow CIs reflect genuine consistency across seeds rather than high precision about population-level effects; we report them for transparency but caution against over-interpreting their width.

我们采用了“报告标志信号”指标，利用 report_flag 分量在槽空间中产生强烈分离（余弦相似度约 0.22 对比约 0.98）的发现。这使我们能够构建出一个针对提示的特定投影，提示存在时为正投影，移位时为负投影。

结果显示提示信号具有容量依赖性：

- 4 个槽（100%）：信号在提示输入时从 0 跳跃到~0.78，经过 4 个掩码步骤保持正值，随后随着槽 0 被覆盖，衰减为负值（~ −0.26）。
- 2 个槽（50%）：初始信号相同（约 0.78），但衰减更快;信号通过掩码步骤 2 变为负（~ −0.41）。
- 总线关闭（0%）：信号始终保持在 0（无工作区存储）。

这种模式类似于人类的反向掩蔽实验：在两种情况下，目标的初始编码在不同条件下（提示输入时的约 0.78 信号）是相同的，但有意识访问的持续时间因容量限制而异。正如更强的掩体或更短的刺激启动非同步（SOA）会减少人类的有意识报告，工作空间容量的减少会缩短信息在我们代理中可访问的窗口。这展示了 GWT 的预测：工作空间

容量决定了任务相关信息在被输入刺激取代前能持续多久。

神经特征追踪能力。满容量时 GBI 平均为 0.66，显示出广泛的跨时段耦合（时隙间隙高度参与），与 GWT 中的“全球广播”直觉相符。在半容量时，GBI 降至 0.03（广播宽度减少;跨时段参与减少），反映出信息共享机会减少，时隙减少。点火锐利度显示预期模式：全功率时 IS≈0.15（n=20 种子间自控 95%置信区间[0.06， 0.26]），半容量时 IS≈0.06（自举 95%置信区间[0.02， 0.10]），主线关闭时 IS = 0（无工作区间无点火）。公交车离开时完全没有点火表明点火不仅与工作区活动相关，还依赖于工作区活动。没有工作空间，就没有 GWT 中有意识访问所体现的非线性放大的基底。

公交审计证实了因果依赖。“总线审计”（见方法）确认总线路径中有强烈梯度（T1 中位雅可比范数~ 78–175），主干线漏气为零（T3 中干隔离 0%翻转），验证了行为确实依赖于工作区内容。

鉴于元认知访问依赖于自我模型，信息访问依赖于能力，我们现在探讨这些指标是否能预测对内部噪声的功能鲁棒性。

5.3 实验 3（三角剖分）：层级鲁棒性和复合标记

5.3.1 动机

意识理论常被视为竞争者，每个理论都提出了独特的神经特征。然而，我们之前的实验表明它们描述的是不同的功能层。我们选择了这个最终角度来测试积分和功能效用维度：将抽象的神经标记（点火、复杂度、校准）与鲁棒性的具体进化优势联系起来。通过弥合这一“解释空白”，我们旨在展示意识的意义。

在实验 3 中，我们检验了一个三角剖分假说：交叉结构标记的简化复合（GBI + ΔPCI）比单独使用这两个术语更能预测鲁棒性，且添加一个 HOT 对齐的元认知项（Type-2 AUROC）能在自建模代理中提供增量信息。我们明确评估这些标记是否捕捉了冗余信息，或是对鲁棒性贡献了独特方差。

5.3.2 选举结果

实验 3 得出了两个相同的结果：（1）限制 PCI 式复杂性代理转移给工程代理的负面结果，（2）区分仅广播架构与高阶监控架构的鲁棒性层级。我们首先介绍 PCI 传输约束，因为它对人工系统中 IIT 相关的评估有直接影响。

[1] 我们注意到，当种子间方差低时，自助信赖区间可能会变得狭窄。这些狭窄的 CI 反映了种子间的真实一致性，而非对种群层面效应的高精度;我们举报它们是为了透明度，但也要谨慎不要过度解读其宽度。

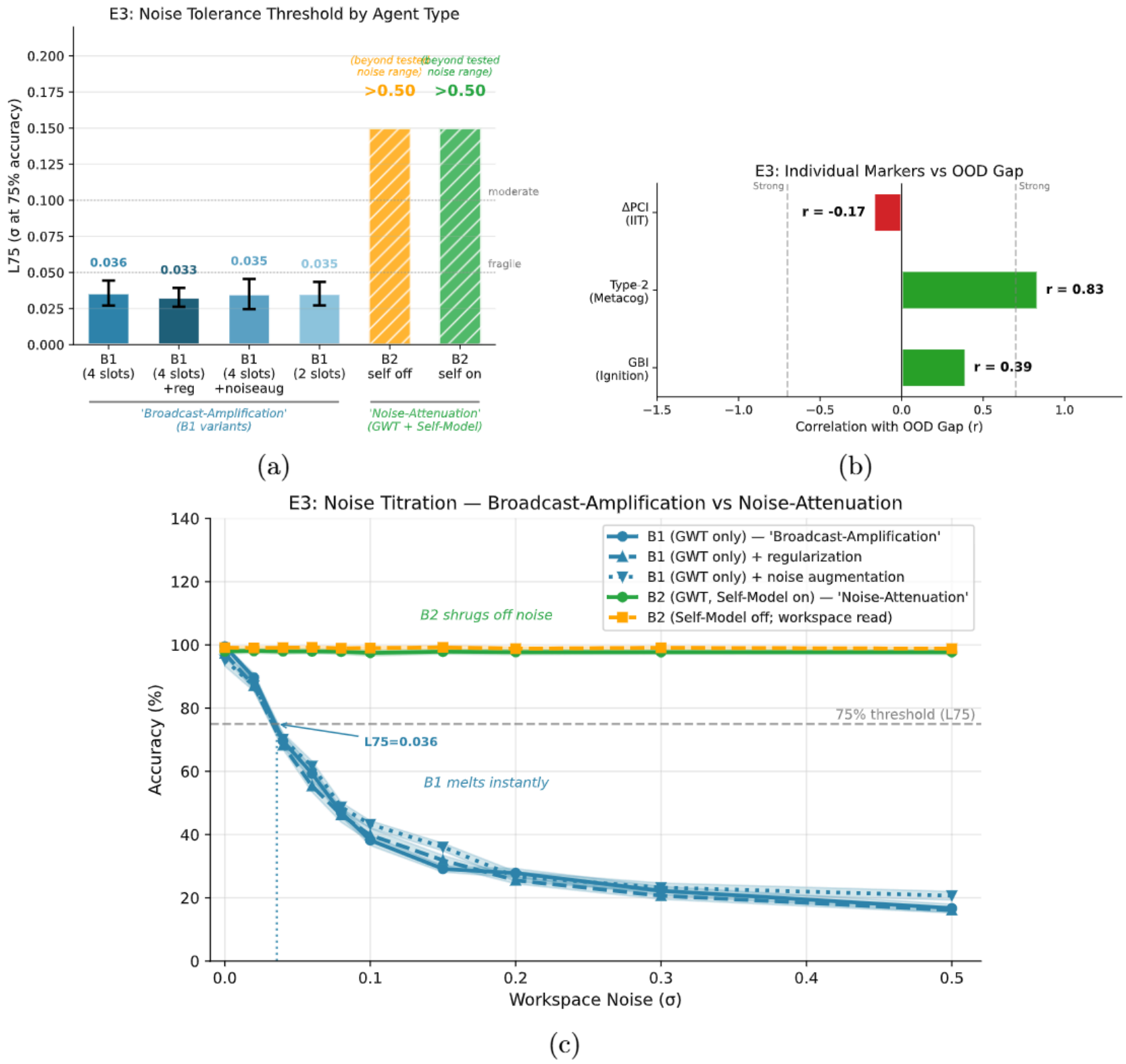

Figure 3: **Two co-equal findings: a PCI transfer constraint and a robustness hierarchy.** (a) **Broadcast-Amplification vs. Noise-Attenuation:** B1 (GWT-only) collapses under minimal latent noise (L75 ≈ 0.04); two steelman B1 variants with standard regularization or training-time noise augmentation remain similarly fragile (L75 ≈ 0.03–0.04). B2 remains robust to latent noise within the tested range (L75 > 0.50, i.e., accuracy never dropped to 75% for $\sigma \leq 0.50$) both with the Self-Model enabled and in a Self-Model-off / workspace-read control. This suggests that, in this implementation, the robustness gain is not solely explained by routing actions through z_{self}. (b) **Marker Correlations with OOD Gap:** Pearson correlations between individual markers and the OOD generalization gap (ID decision accuracy minus OOD decision accuracy; larger gaps indicate worse robustness). For agents without a workspace (and for the bus-off lesion), we define GBI as 0 ("no broadcast") for this correlation. The IIT-adjacent term is ΔPCI (correct minus incorrect PCI-A; Methods), not raw PCI-A; raw PCI-A is reported separately and is inverted under the workspace bottleneck (Table 3). (c) **Noise Titration Curves:** Performance degradation profiles for baselines, GWT (fragile), and Self-Model (robust) agents under latent perturbation.

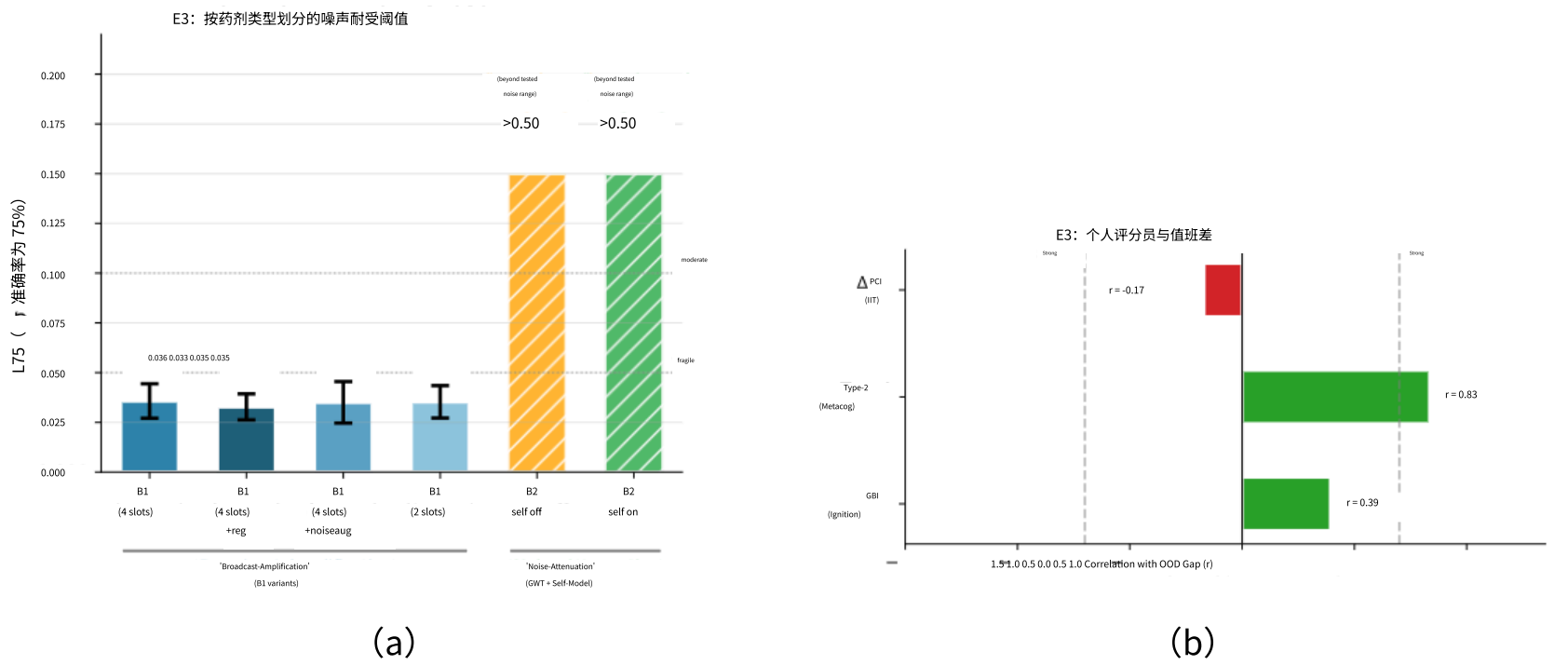

图 3：两个相同的发现：PCI 传输约束和鲁棒性层级。（a） 广播放大与噪声衰减：B1（仅 GWT）在极小情况下坍缩潜噪声（L75 ≈ 0.04）;两款带有标准正则化或训练时间噪声增强的 Steelman B1 变体依然脆弱（L75 ≈ 0.03–0.04）。在启用自建模或自建模关闭/工作区读取控制时，B2 在测试范围内（L75>0.50，即σ ⩽0.50 的准确率从未降至 75%），保持对潜在噪声的韧性。这表明，在本实现中，鲁棒性的提升不仅仅由通过 z 的动作路由来解释。（b）标记与 OOD 差距的相关性：单个标记与 OOD 泛化差距之间的 Pearson 相关性（ID 决策准确率减去 OOD 决策准确性;差距越大，鲁棒性越差）。对于没有工作空间的代理（以及公交线断裂病变），我们定义 GBI 为 0（“无广播”），以表其相关性。IIT 的相关术语是 ΔPCI（正确但不正确，PCI-A;方法），而不是原始的 PCI-A;原始 PCI-A 单独报告，并倒置在工作区瓶颈下（见表 3）。（c） 噪声滴定曲线：基线、GWT（脆弱）和自模型（稳健）代理在潜在扰动下的性能退化曲线。

Table 3: **Raw PCI-A is inverted under a workspace bottleneck.** Mean ± SD across $n = 20$ seeds. Higher values indicate higher Lempel–Ziv compressibility-based trajectory complexity.

Architecture	**Raw PCI-A**
A0 (Feedforward baseline)	0.84 ± 0.18
A1 (Recurrent baseline)	0.78 ± 0.02
B1 (Workspace / GWT agent)	0.33 ± 0.02

Finding 1: Raw PCI-A Inverts Under a Workspace Bottleneck: A Transfer Constraint for IIT-Adjacent Proxies. Table 3 reports the *raw* perturbational complexity (PCI-A; see Methods) across baseline and workspace agents. Contrary to a naive IIT/PCI transfer expectation ("more integrated" ⇒ higher perturbation-evoked trajectory complexity), the workspace agent showed *lower* algorithmic complexity than both baselines (A0/A1). This is not a small or ambiguous effect; it is an explicit inversion that functions as a genuine empirical constraint on how PCI-style proxies behave in engineered systems.

Why does the workspace reduce trajectory complexity? Mechanistically, our global workspace is a shared bottleneck that many downstream computations depend on. That design can reduce apparent complexity in at least three (non-exclusive) ways. (i) *Low-rank forcing / redundancy:* if many units are driven by a common broadcast state, the evoked trajectories become more redundant across dimensions and therefore more compressible. (ii) *Manifold stabilization:* bottlenecks and normalization can push dynamics toward a small set of task-relevant attractor-like modes, so perturbations decay back toward a structured manifold rather than producing richly branching transients. (iii) *Failure-driven "complexity" in baselines:* recurrent baselines can produce high Lempel–Ziv complexity through unstructured variability, especially when the perturbation pushes them off-policy; high compressibility-based complexity is not guaranteed to reflect meaningful integration.

Is this a fundamental problem with PCI as a consciousness measure? We interpret the result as *implementation- and setting-specific*, not as a refutation of PCI or IIT. PCI in humans is designed around spatiotemporal complexity across many channels following a perturbation (e.g., TMS–EEG) in a system whose baseline dynamics include rich spontaneous activity. Our PCI-A proxy instead measures the compressibility of a particular internal trajectory under an engineered bottleneck and a synthetic perturbation scheme. The negative result therefore primarily cautions that "absolute algorithmic complexity of a latent trajectory" is not an architecture-invariant stand-in for integration when moved from biological cortex to task-trained artificial agents.

Why use ΔPCI at all? We report ΔPCI (correct minus incorrect trial complexity) as a contrastive *selective complexity* diagnostic that partially controls for noise/failure confounds. In these agents, incorrect trials often reflect destabilized, off-policy behavior that can inflate compressibility-based complexity without corresponding to successful information processing. Conditioning on correctness treats behavioral success as an external anchor for "meaningful" processing and asks a narrower question: does perturbational complexity increase *selectively* when the agent is in the regime that supports correct computation? We emphasize that ΔPCI is not a canonical IIT prediction and should not be read as a principled surrogate for Φ; it is a pragmatic control that helps separate *structured* evoked dynamics from *unstructured* failure-driven variability. In our hierarchical regression, its incremental contribution to OOD robustness is modest ($\Delta R^2 \approx 2\%$), and we interpret this as further evidence that PCI-style proxies require care when ported to engineered agents.

Finding 2: Broadcast-Amplification vs. Noise-Attenuation. The noise titration revealed a dissociation between broadcast capacity and robustness (Figure 3a). The GWT agent (B1), despite having high report-step decision accuracy (100%) and high GBI (0.65), proved fragile, exhibiting broadcast-amplification with an L75 of $\sigma \approx 0.04$. To put this in perspective: the workspace activations have unit variance after normalization, so $\sigma = 0.04$ represents noise at just 4% of the signal magnitude. Yet this tiny perturbation was sufficient to drop performance from 100% to 75%. By $\sigma = 0.10$ (10% noise), accuracy had collapsed to near chance.

To address the concern that this is a straw-man baseline, we repeated the titration on B1 with two standard robustness interventions: (i) dropout + weight decay during behavior cloning, and (ii) training-time latent workspace-noise augmentation. Neither materially improved noise tolerance (L75 $= 0.033 \pm 0.007$ and 0.035 ± 0.010, respectively; $n = 20$ seeds).

This extreme sensitivity arises because B1's policy reads directly from the workspace slots. Any noise injected into the workspace propagates directly to the action selection, with no opportunity for filtering

表 3：原始 PCI-A 在工作区瓶颈下被反转。平均 ± SD 跨 n = 20 个种子。
值越高，表示基于 Lempel–Ziv 可压缩性的轨迹复杂度越高。

建筑	原始 PCI-A
A0（前馈基线）0.84 ± 0.18 A1（循环基线）0.78 ± 0.02 B1（工作空间/GWT 代理）0.33 ± 0.02	

发现 1：原始 PCI-A 在工作区瓶颈下反转：传输约束
IIT 邻近代理。表 3 报告了基线和工作区代理的原始扰动复杂度（PCI-A;见方法）。与简单的 IIT/PCI 传输预期（“更集成”⇒更高受扰动引发的轨迹复杂度）相反，工作区代理的算法复杂度低于两个基线（A0/A1）。这不是小或模糊的影响;它是一种显式的反转，作为对 PCI 式代理在工程系统中行为的真实经验约束。

为什么工作区能降低路径复杂度？从机制上讲，我们的全球工作空间
是许多下游计算依赖的共享瓶颈。该设计至少可以通过三种（非排他）方式降低表面复杂性。（i） 低秩强迫/冗余：如果许多单元由共同广播态驱动，诱发的轨迹在不同维度上变得更冗余，因此更具可压缩性。（ii） 流形稳定：瓶颈和归一化会将动力学推向一小部分与任务相关的吸引子样模态，因此扰动会衰减回结构化流形，而非产生丰富分支的瞬态。（iii）基线中的失效驱动“复杂性”：循环基线通过无结构变异产生高 Lempel–Ziv 复杂性，尤其是在扰动使其偏离策略时;基于高可压缩性的复杂度并不保证能反映有意义的积分。

这是 PCI 作为意识测量的一个根本性问题吗？我们对结果进行解释
作为实现和环境特定，而非对 PCI 或 IIT 的反驳。人类的 PCI 设计基于在系统中基线动态丰富的自发活动（如 TMS–EEG）扰动后，跨多个通道实现时空复杂性。我们的 PCI-A 代理则测量特定内部轨迹在工程瓶颈和合成微扰方案下的可压缩性。因此，否定结果主要提醒人们，“潜在轨迹的绝对算法复杂度”并非从生物皮层转移到任务训练人工代理时的架构不变整合替代。

为什么还要用ΔPCI？我们报告ΔPCI（正确减去错误试验复杂度）作为对比选择性复杂度诊断，部分控制噪声/失效混杂因素。在这些代理中，错误的试验往往反映了不稳定、偏离策略的行为，这些行为可能使基于可压缩性的复杂性膨胀，却未能实现信息处理的成功。对正确性的条件将行为成功视为“有意义”处理的外部锚点，并提出一个更狭窄的问题：当主体处于支持正确计算的状态时，扰动复杂性是否会选择性增加？我们强调ΔPCI 并非 IIT 的典范预测，不应被视为Φ的原则性替代;这是一种务实控制，帮助区分结构化的诱发动力学与非结构化的失效驱动变异。在我们的层级回归中，它对 OOD 鲁棒性的增量贡献有限（ΔR≈ 2%），我们将其解读为进一步证明 PCI 风格代理在移植到工程代理时需要谨慎。

发现 2：广播放大与噪声衰减。噪声滴定显示广播容量与鲁棒性之间存在解离（图 3a）。尽管 GWT 代理（B1）具有高报告步骤决策准确率（100%）和高 GBI（0.65），但表现脆弱，广播放大表现为 L75 σ ≈0.04。举个例子：工作区激活在归一化后有单位方差，因此 σ = 0.04 代表仅 4% 信号幅度的噪声。然而，这一微小的扰动足以使性能从 100%降至 75%。到了σ = 0.10（10%噪声），准确率已降至接近偶然。

为解决该基线问题的担忧，我们在 B1 上重复了两次标准稳健性干预的滴定：（i）行为克隆过程中的退出+体重下降，以及（ii）训练时间潜在工作区噪声增强。两者都没有实质性提升噪声耐受性（L75 = 0.033 ± 0.007 和 0.035 ± 0.010;n = 20 个种子）。

这种极高的敏感性是因为 B1 的策略直接读取工作区槽。任何注入工作区的噪声都会直接传播到动作选择中，没有过滤的机会

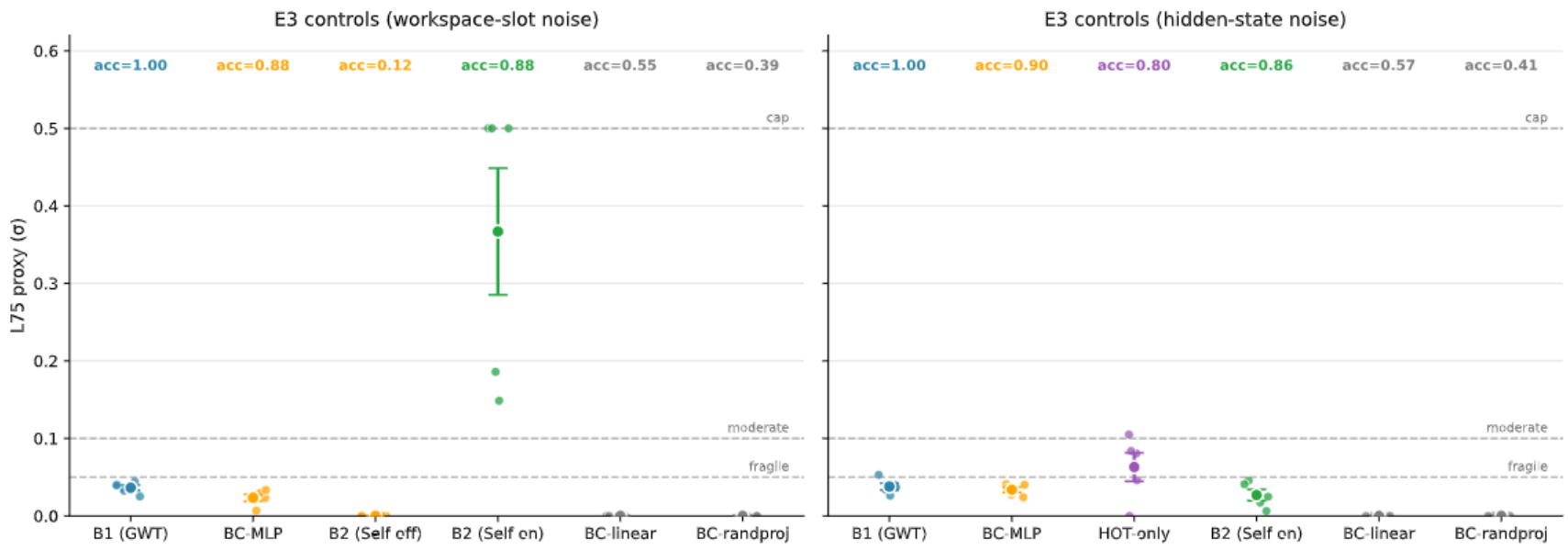

Figure 4: **Denoising attribution controls (pilot; $n = 5$ seeds).** Left: latent workspace-slot noise (the primary Experiment 3 perturbation) comparing B1 (GWT-only), B2 (Self-Model), and matched "BC-only" compressor controls (a 64-d bottleneck applied directly to the flattened workspace state). Right: latent hidden-state (post-GRU carrier) noise comparing the same workspace agents to a "HOT-only" agent (A1 + z_{self} without workspace; cues visible because there is no workspace bus). Points are seeds; squares show mean ± SD for finite thresholds; triangles at $\sigma = 0.5$ denote seeds that did not cross the 75% threshold within the tested noise range.

or error correction. The workspace acts as an amplifier: it broadcasts information globally, but it also broadcasts noise globally.

In contrast, the Self-Model agent (B2) demonstrated robustness beyond the tested noise range (L75 > 0.50; accuracy never dropped to 75% for $\sigma \leq 0.50$), exhibiting noise-attenuation. Even at $\sigma = 0.50$ (noise equal to 50% of signal magnitude), B2 maintained >80% accuracy. This robustness persisted in the Self-Model-off / workspace-read control (Figure 3a), indicating that in this implementation robustness is not strictly dependent on routing actions through z_{self} at inference. While z_{self} is a strongly compressed summary of the workspace state (effective dimension ∼2), the noise-tolerance advantage appears to be a broader property of the B2 training/architecture rather than a bottleneck-only effect.

Denoising attribution baselines. A key alternative hypothesis is that the robustness gain in Figure 3 is not specific to the Self-Model, but would emerge from adding any low-dimensional bottleneck on the action pathway. To test this, we evaluated (i) a "HOT-only" agent (A1 + z_{self} without a workspace bus) and (ii) matched "BC-only" compressor controls that replace z_{self} with a 64-d bottleneck on the flattened workspace state. In a pilot sweep ($n = 5$ seeds; Figure 4), the learned MLP bottleneck achieves high baseline accuracy (0.879 ± 0.054) but remains fragile under workspace-slot noise (L75 $= 0.023 \pm 0.010$), far below the Self-Model agent (3/5 seeds remain above the 75% threshold at $\sigma = 0.5$, and the remaining seeds yield L75 $= 0.167 \pm 0.026$). Linear and random-projection bottlenecks underperform on baseline accuracy and do not yield meaningful robustness comparisons in this task. Under hidden-state noise, the HOT-only agent (evaluated with cues visible, since it has no workspace bus) maintains baseline accuracy 0.799 ± 0.169 and shows improved tolerance (L75 $= 0.063 \pm 0.041$), indicating that a Self-Model bottleneck can also stabilize recurrence-only computation, but that the strongest robustness effect in this study remains specific to the workspace-noise regime.

Composite Predicts OOD Fragility Across Architectures. In the canonical OOD-gap analysis (Figure 3b), we correlate individual markers with the OOD generalization gap (ID accuracy minus OOD accuracy; larger gaps indicate worse generalization). The *positive* correlations indicate that higher marker values are associated with *larger* OOD gaps (i.e., worse OOD generalization). This is distinct from L75 noise tolerance, where B2 excels; OOD fragility reflects a different failure mode (distribution shift rather than latent noise). Broadcasting alone has modest predictive power: GBI yields Pearson $r \approx 0.39$ (equivalently $R^2 \approx 0.15$). Adding ΔPCI provides only a small incremental gain ($\Delta R^2 \approx 2\%$). Adding Type-2 AUROC improves in-sample fit (Pearson $r \approx 0.79$; leave-one-out $r_{\text{LOO}} \approx 0.62$), but AUROC is near-constant for non–Self-Model agents, so this term partly encodes the Self-Model design choice.

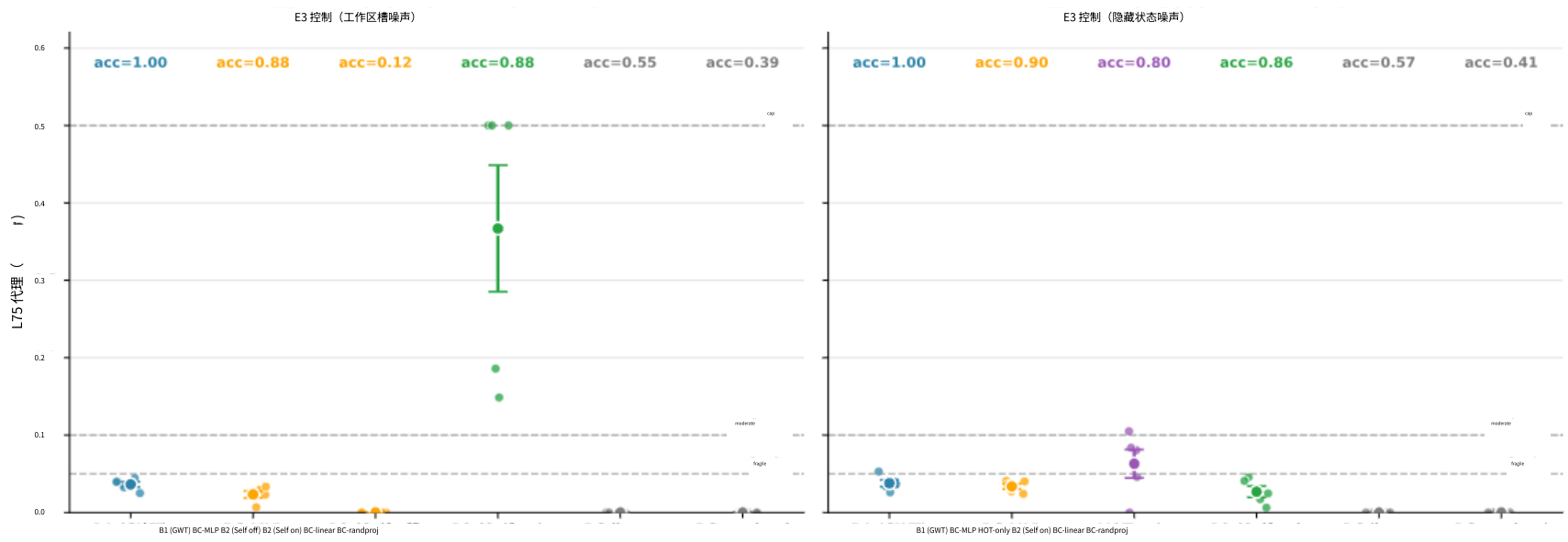

图 4：去噪归因控制（试点;n = 5 个种子）。左图：潜在工作区槽噪声（实验 3 的主要扰动），比较 B1（仅 GWT）、B2（自建模）和匹配的“仅 BC 限”压缩器控制（直接应用于平面工作区状态的 64 维瓶颈）。右：潜在隐藏状态（后 GRU 载体）噪声，将相同工作区代理与“仅 HOT”代理（A1，无过滤机会+zwithout workspace;因无工作区总线可见提示）进行比较。积分是种子;平方表示有限阈值的均值±标准差;三角形 σ = 0.5 表示未在测试噪声范围内超过 75% 阈值的种子。

或者纠错。工作空间充当放大器：它不仅全球广播信息，也在全球范围内广播噪声。

相比之下，自建模代理（B2）在测试噪声范围（L75 0.50;准确率从未降至 75%σ ⩽0.50），并表现出噪声衰减。即使在σ = 0.50（噪声等于信号幅度的 50%）下，B2 仍保持>80%的准确率。这种鲁棒性在自建模关闭/工作区读取控制中依然存在（图 3a），表明在该实现中鲁棒性并不严格依赖于通过 zat 推理路由操作。虽然这是工作空间状态（有效维数为 ~2）的强烈压缩总结，但噪声耐受优势似乎是 B2 训练/架构的更广泛属性，而非仅限于瓶颈效应。

去噪归因基线。一个关键的替代假说是，图 3 中的鲁棒性增益并非自模型特有，而是通过在作用路径上添加任何低维瓶颈而产生的。为此，我们评估了（i）一个“仅 HOT”代理（A1 + z 但无工作区总线）和（ii）匹配的“仅限 BC”压缩控制，这些控制用 64 维瓶颈替换了 z，位于平坦工作区状态。在试点扫描中（n = 5 个种子;图 4），所学的 MLP 瓶颈实现了较高的基线准确率（0.879 ± 0.054），但在工作区槽噪声（L75 = 0.023 ± 0.010）下仍然脆弱，远低于自建模代理（3/5 种子在σ = 0.5 时仍超过 75%阈值，剩余种子产出 L75 = 0.167±0.026）。线性和随机投影瓶颈在基线准确性上表现不佳，且在此任务中无法产生有意义的鲁棒性比较。在隐藏状态噪声下，仅 HOT 的代理（由于没有工作区总线，评估时有可见线索）基线准确率为 0.799±0.169，并表现出更好的容忍度（L75 = 0.063 ± 0.041），表明自建模瓶颈也能稳定仅重现计算，但本研究中最强的鲁棒性效应仍特有于工作区噪声区。

Composite 预测不同架构的 OOD 脆弱性。在规范的值班缺口分析中（图 3b），我们将个别标志物与 OOD 泛化差距（ID 准确率减去 OOD 准确率;差距越大表示泛化越差）。正相关性表明，较高的标记值与更大的离场差距相关（即更差的离场泛化）。这与 L75 的噪声容忍度不同，后者 B2 更为出色;OOD 脆弱性反映了不同的失效模式（分布偏移而非潜在噪声）。仅广播预测力有限：GBI 给出皮尔逊 r≈0.39（相当于 R≈0.15）。加入ΔPCI 仅带来小幅增益（ΔR≈ 2%）。添加 2 型 AUROC 可以改善样本内拟合度（Pearson r ≈ 0.79;省略一 r≈ 0.62），但对于非自建模代理，AUROC 几乎是恒定的，因此该术语部分编码了自建模设计选择。

Non-Circularity Controls. Because Type-2 AUROC is ≈ 0.50 for agents lacking a Self-Model, we treat AUROC-free composites (GBI + ΔPCI) as the primary cross-architecture triangulation score and interpret AUROC gains as mechanism-specific rather than independent evidence. Accordingly, we treat the CTS→L75 link as a *mechanistic consistency check* within this synthetic setting rather than a discovery-grade predictive model; broader generalization would require additional architectures where AUROC varies continuously.

Unique Variance Decomposition. Hierarchical regression (Figure 3c) revealed the distinct contributions of each layer. In hierarchical regression, predictors are added sequentially to determine how much additional variance each explains beyond what is already captured by earlier predictors. The R^2 value indicates the proportion of variance explained (ranging from 0 to 1, where 1 means perfect prediction), and ΔR^2 indicates the unique contribution of each added predictor.

- **Layer 1 (Broadcast):** GBI alone explains ∼15% of the variance in OOD gap (Pearson $r \approx 0.39$). This confirms that broadcasting capacity matters, but leaves ∼85% unexplained.
- **Layer 2 (Integration):** Adding ΔPCI yields a small but non-zero gain ($\Delta R^2 \approx 2\%$), bringing the total to ∼17%. This does *not* overturn the inverted absolute PCI-A ordering (Table 3); instead it reframes the question contrastively: does perturbational complexity increase *selectively* on successful trials ("meaningful processing") relative to failed trials ("noise complexity")?
- **Layer 3 (Metacognition):** Adding Type-2 AUROC yields a further gain ($\Delta R^2 \approx 19\%$), bringing the total to ∼37%. However, as noted above, AUROC is near-constant (0.50) for all agents lacking a Self-Model, so this term primarily distinguishes B2 self-on from the rest. The key result is that **GBI + ΔPCI alone explain ∼17% of variance in OOD gap**, markers that are well-defined across all architectures.

This pattern is consistent with a hierarchical interpretation: broadcast capacity (GBI) and processing richness (ΔPCI) provide the foundation, while Type-2 AUROC mainly distinguishes Self-Model agents; the robustness control in Figure 3a suggests the robustness gain is not solely mediated by z_{self} at inference. We caution against interpreting the AUROC contribution as independent evidence, given its structural confound with the Self-Model design.

6 General Discussion

6.1 Summary of Findings

We introduced a synthetic neuro-phenomenology framework to stress-test consciousness indicators. Our experiments reveal a hierarchical functional architecture defined by both capabilities and surprising failure modes:

Experiment 1 showed a double dissociation in this agent family: Self-Model ablation abolishes metacognitive calibration ("knowing") while leaving first-order performance ("doing") largely intact.

Experiment 2 provides evidence that workspace capacity is causally necessary for information access within this setup ($n = 20$ seeds; full vs. half capacity: $p < 10^{-4}$, $g = 1.65$; B1 vs. A0/A1: $p < 10^{-9}$, $g > 3$), and reveals a sharp discontinuity at the complete workspace lesion (bus-off) across the tested discrete capacity settings. When the workspace is fully lesioned (bus-off), the access-related dynamics collapse entirely; with $n = 20$ seeds and discrete capacity levels, we treat this as a qualitative regime contrast rather than a precisely estimated threshold.

Experiment 3 revealed the **broadcast-amplification** effect: the global workspace broadcasts information but amplifies noise, creating extreme fragility. B2 agents remain robust in the same perturbation regime (including in a Self-Model-off / workspace-read control), yielding a **noise-attenuation** pattern relative to B1. Experiment 3 also produced an explicit negative result for a naive IIT/PCI-style complexity proxy: raw PCI-A is inverted under the workspace bottleneck, and ΔPCI contributes only modest unique variance.

6.2 Indicators vs. Pseudo-Consciousness

Our agents are not conscious; they are reference implementations of theories. The fact that B2 passes HOT-style Self-Model tests and B1 passes GWT tests without "feeling" anything highlights the risk

非圆形控制。由于 2 型 AUROC 对缺乏自模型的代理者≈为 0.50，我们将无 AUROC 的复合指标（GBI + ΔPCI）视为主要跨架构三角剖分，并将 AUROC 增益视为机制特异性证据，而非独立证据。因此，我们将 CTS→L75 链接视为该综合环境中的机制性一致性检查，而非发现级预测模型;更广泛的推广则需要额外的架构，使 AUROC 持续变化。

唯一方差分解。层级回归（图 3c）揭示了每层的独特贡献。在层次回归中，预测变量会依次添加，以确定每个变量能解释超出先前预测变量已涵盖的额外方差。R 值表示解释的方差比例（范围为 0 到 1,1 表示完美预测），Δ表示每个新增预测变量的唯一贡献。

- 第一层（广播）：仅 GBI 就能解释约 15%的值班时间差距方差（Pearson r ≈ 0.39）。
  这证实了广播容量的重要性，但留下了约 85%的解释。
- 第二层（集成）：加入ΔPCI 可获得小幅但不为零的增益（ΔR≈ 2%），总增益约为 17%。这并不推翻倒置的绝对 PCI-A 排序（见表 3）;相反，它以对比的方式重新表述了这个问题：相对于失败试验（“噪声复杂度”），扰动复杂度是否会选择性增加（“有意义处理”）？
- 第三层（元认知）：加入第二类 AUROC 可获得进一步收益（ΔR≈ 19%），总收益达到约 37%。然而，如上所述，AUROC 对于所有没有自模型的代理人几乎保持恒定（0.50），因此该术语主要区分 B2 自启动与其他代理。关键结果是，仅 GBI + ΔPCI 就能解释约 17%的 OOD 差距方差，这些标志在所有架构中都定义得很明确。

这一模式符合层级解释：广播容量（GBI）和处理丰富度（ΔPCI）为基础，而 Type-2 AUROC 主要区分自建模代理;图 3a 中的鲁棒性控制表明，鲁棒性提升并非仅由 ZAT 推断介导。鉴于 AUROC 与自模型设计的结构性混淆，我们谨慎避免将其视为独立证据。

6 综合讨论

6.1 调查结果总结

我们引入了综合神经现象学框架，用于对意识指标进行压力测试。我们的实验揭示了一种由能力和令人惊讶的失效模式共同定义的层级功能架构：实验 1 显示了该代理家族中的双重解离：自建模消融消除了元认知校准（“知道”），同时基本保留了一阶性能（“执行”）。

实验 2 提供了证据，表明在该设置中，工作区容量是信息访问的因果必要条件（n = 20 个种子;满容量与半容量：p <10，g = 1.65;B1 与 A0/A1：p < 10，g > 3），在被测试的离散容量设置中，完全工作区病变处（总线关闭）处出现了明显的不连续性。当工作空间完全被破坏（公交线关闭）时，与访问相关的动态完全崩溃;当 n=20 个种子且产能水平不同时，我们将其视为定性对比，而非精确估计的阈值。

实验 3 揭示了广播放大效应：全球工作空间广播信息但放大噪声，造成极度脆弱。B2 代理在相同的扰动区间（包括自建模/工作区读取控制）下保持稳健，相对于 B1 具有噪声衰减模式。实验 3 还对一个简单的 IIT/PCI 风格复杂性代理产生了明确的负结果：原始 PCI-A 在工作区瓶颈下被反转，ΔPCI 仅贡献适度的唯一方差。

6.2 指标与伪意识

我们的代理人是没有意识的;它们是理论的参考实现。B2 通过了 HOT 风格的自建模测试，而 B1 通过了 GWT 测试却没有“感受”任何东西，这凸显了风险

of pseudo-consciousness. However, the *functional* properties revealed (especially the robustness of the B2 family under latent noise perturbations) suggest that these architectural features provide tangible evolutionary advantages beyond mere "inner light."

6.3 Hierarchies of Robustness and their Limits

The dissociation between B1 (broadcast-amplification) and B2 (noise-attenuation) challenges the assumption that broadcasting equals robustness. The global workspace amplifies signals, but without a higher-order monitor, it also amplifies noise. This suggests that GWT and Self-Model mechanisms co-evolved: workspace for rapid dissemination, Self-Model for quality control.

However, the Self-Model is not a "homunculus" that cares about truth; it is a mechanism optimized for reward. This implies that introspective access is not an automatic byproduct of having a Self-Model; it emerges from the architectural constraint that forces behavior to be mediated by the compressed self-representation.

6.4 A Constraint on PCI-Style Transfer to Engineered Agents

The inversion of raw PCI-A under the workspace bottleneck (Table 3) is one of the most consequential findings for the IIT community. Naive transfer of PCI-style measures (where "more integrated" architectures are expected to show higher perturbation-evoked trajectory complexity) fails in our engineered agents. This is not a methodological nuisance but a genuine empirical constraint: in systems with shared bottlenecks and normalization, tighter coupling can *reduce* apparent complexity by forcing dynamics onto lower-dimensional manifolds and increasing redundancy across channels. Meanwhile, unstructured failure dynamics in less-organized systems can inflate Lempel–Ziv complexity without reflecting meaningful integration.

This result does not refute PCI or IIT as applied to biological systems, where the measure was designed for spatiotemporal complexity across many channels following TMS perturbations in a brain with rich spontaneous activity. Rather, it cautions that absolute algorithmic complexity of a latent trajectory is not architecture-invariant when moved from biological cortex to task-trained artificial agents. Future work on AI consciousness assessment should explicitly validate which perturbation and readout choices recover IIT-aligned monotonicities before treating PCI-style proxies as universal integration measures.

6.5 Implications for AI Assessment

Our results support a triangulation approach to AI consciousness assessment: no single marker is sufficient, and different markers can fail in different ways. In our synthetic setting, a reduced composite of cross-architecture markers (GBI + ΔPCI) predicts robustness, while metacognitive calibration (Type-2 AUROC) provides additional diagnostic power primarily *within* Self-Model agents. This avoids treating AUROC as a universal continuous predictor in a regime where it is near-constant for non–Self-Model architectures. The core functional lesson remains: a system that ignites and broadcasts without quality control can be brittle (B1), whereas a system that filters and monitors its own broadcast stream is robust (B2).

This has practical implications for AI safety and alignment. An AI system that lacks metacognitive calibration cannot accurately report its own uncertainty, making it difficult for human operators to know when to trust its outputs. Our results suggest that metacognitive capacity is not merely a philosophical curiosity but a functional requirement for reliable AI systems. Future assessments should prioritize such multi-theory triangulation over the search for single canonical indicators, asking not just "does this system broadcast information?" but also "does this system know what it knows?"

6.6 Limitations and Future Work

Several limitations should be noted. First, we report statistics across $n = 20$ training seeds per condition for the main comparisons in Experiments 1–3 (seed as the unit of analysis). Some targeted sweeps and attribution controls are pilots ($n = 5$ seeds) and should be interpreted accordingly. This sample size is sufficient to demonstrate large qualitative regime changes, but it remains underpowered for moderate effects and yields uncertainty around estimated thresholds; wide intervals in intermediate regimes (e.g., reduced capacity) reflect genuine seed-to-seed variability rather than mere measurement noise. We did not run a prospective power analysis, and we treat quantitative effect sizes and estimated thresholds as exploratory proof-of-concept results within this specific architecture family and task suite.

任何“感觉”都凸显了伪意识的风险。然而，揭示的功能特性（尤其是 B2 家族在潜在噪声扰动下的鲁棒性）表明，这些结构特征提供了超越单纯“内在光”的具体进化优势。

6.3 稳健性层级及其限制

B1（广播放大）和 B2（噪声衰减）之间的解离挑战了广播等同于鲁棒性的假设。全球工作区放大信号，但没有高阶监视器，也会放大噪声。这表明全球工作技术（GWT）和自建模机制是共同进化的：工作空间用于快速传播，自建模用于质量控制。

然而，自我模型并不是关心真理的“小人”;这是一种为奖励优化的机制。这意味着内省访问并非拥有自我模型的自动副产品;它源于一种结构约束，迫使行为必须通过压缩的自我表征来中介。

6.4 PCI 风格传输到工程代理的限制

原始 PCI-A 在工作空间瓶颈下的反转（见表 3）是 IIT 社区最重要的发现之一。在我们的工程代理中，简单地转移 PCI 式度量（“更集成”的架构预期表现出更高的微扰引发轨迹复杂性）在我们的工程智能体中失败了。这不是方法论上的麻烦，而是真实的经验约束：在共享瓶颈和归一化的系统中，更紧密耦合可以通过将动态推向低维流形并增加通道间冗余来降低表观复杂度。与此同时，在组织较少的系统中，无结构的失效动力学可能会膨胀 Lempel–Ziv 复杂性，却无法反映有意义的积分。

这一结果并不反驳 PCI 或 IIT 在生物系统中的应用，后者在具有丰富自发活动的大脑中，针对 TMS 扰动后，跨多个通道实现时空复杂性。相反，它提醒人们，潜在轨迹的绝对算法复杂度在从生物皮层转移到任务训练的人工代理时，并非建筑结构不变。未来的 AI 意识评估研究应明确验证哪些扰动和读出选择恢复了 IIT 对齐的单调性，然后再将 PCI 式代理作为通用积分指标。

6.5 对人工智能评估的影响

我们的结果支持一种三角测量方法来评估人工智能意识：没有单一标志是足够的，不同标记可能以不同方式失效。在我们的综合环境中，跨架构标记的简化复合（GBI + ΔPCI）预测鲁棒性，而元认知校准（Type-2 AUROC）主要在自建模代理中提供额外的诊断能力。这避免了在非自模型架构中 AUROC 近乎恒定的环境中，将 AUROC 视为通用连续预测变量。核心功能教训依然存在：一个在没有质量控制的情况下点燃和广播的系统可能很脆弱（B1），而一个能过滤和监控自身广播流的系统则是稳健的（B2）。

这对人工智能的安全性和一致性具有实际意义。缺乏元认知校验的人工智能系统无法准确报告自身的不确定性，这使得人类操作员难以判断何时该信任其输出。我们的结果表明，元认知能力不仅仅是哲学上的好奇心，而是可靠人工智能系统的功能性需求。未来的评估应优先考虑这种多理论三角定位，而非单一典型指标的搜索，不仅要问“这个系统是否广播信息？”，还要问“这个系统是否知道它所知道的？”

6.6 局限性与未来工作

需要注意几个限制。首先，我们报告实验 1–3 中主要比较中 n=20 个训练种子的统计数据（以种子为分析单位）。一些有针对性的扫除和归因控制是试点（n=5 个种子），应相应解读。该样本量足以证明较大的定性状态变化，但对于中等效应仍不足，且在估计阈值附近存在不确定性;中间区间宽（如容量降低）反映的是种子间真实的变异性，而非单纯的测量噪声。我们未进行前瞻性功效分析，并将定量效应大小和估计阈值视为该特定架构家族和任务组内的探索性概念验证结果。

Second, our IIT-adjacent complexity probe produced an explicit negative result for a naive PCI-style transfer prediction: raw PCI-A was inverted for workspace agents (Table 3). We argue this inversion is informative rather than embarrassing: in engineered bottlenecked systems, tighter coupling and organization can reduce compressibility-based trajectory complexity, while unstructured failure dynamics can inflate it. While ΔPCI provides a modest contrastive "selective complexity" signal, it remains a heuristic diagnostic (not a direct test of Φ) whose behavior depends on how failure/noise contributes to trajectories. Future work should implement more direct approximations to Φ and/or alternative whole-vs-parts integration measures that are better matched to software agents, and should explicitly test which perturbation and readout choices recover the intended IIT-aligned monotonicities.

Third, our tasks were relatively simple gridworlds with discrete actions and limited state spaces. The broadcast-amplification / noise-attenuation distinction may manifest differently in richer, continuous-control environments where noise can accumulate over longer time horizons.

Denoising attribution controls (scope). We include HOT-only and BC-only compressor controls (Figure 4) to tighten attribution for the Experiment 3 robustness result. However, these controls were run as a pilot ($n = 5$ seeds) and span a limited compressor family (linear projection, small MLP, fixed random projection). Some compressor baselines also underperform on baseline accuracy, which limits the interpretability of their robustness thresholds. Future work should expand these attribution controls (more seeds; parameter-matched variants; additional bottleneck placements; and stronger learned compressors such as autoencoder-style reconstruction objectives) to test whether conclusions remain stable under broader architectural variation.

Fourth, our agents are trained via behavior cloning from an expert oracle, and Experiment 1 includes an explicit auxiliary loss supervising the confidence head. This raises a potential concern that the Self-Model could merely learn to predict the oracle's behavior rather than monitor the agent's own epistemic state. Two clarifications are important. (i) The oracle is used only to generate training targets; it is never consulted at inference, and all reported effects are measured in autonomous rollouts of trained agents under *post-training* perturbations (lesions, capacity reduction, and latent workspace noise) that are out-of-distribution relative to the demonstrations. (ii) The metacognitive supervision target is defined by the agent's own intended report decision being correct (argmax of the agent logits compared to the oracle-provided ground-truth label at that step). Thus, the Self-Model is trained to answer "will *my* current decision be correct given my internal state?" rather than to predict whether the oracle would be correct. Nevertheless, reinforcement learning can interact with the same architectures in qualitatively different ways (exploration, credit assignment, and stability), and establishing whether comparable dissociations emerge under RL (particularly whether calibrated metacognition can be learned from reward alone without labeled correctness) remains an open and important direction for future work.

Finally, our agents are not conscious in any phenomenological sense; they are reference implementations of theories. The fact that B2 passes HOT tests without "feeling" anything underscores the gap between functional indicators and subjective experience. However, we argue that functional validation is a necessary (if not sufficient) step toward understanding consciousness: if a proposed mechanism does not produce the predicted functional consequences even in a simplified synthetic system, it is unlikely to be the correct account of the biological phenomenon.

Future work will investigate how these architectural components interact under greater temporal conflict, test whether the hierarchical relationship between broadcast and metacognition generalizes to language models and other modern AI architectures, and explore mechanisms for maintaining metacognitive stability without explicit supervision. Ultimately, the goal is to establish whether the triad of broadcast, integration, and metacognition serves as a robust predictor of general intelligence and adaptability across diverse domains.

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其次，我们对 IIT 的邻近复杂度探针给出了一个显式的负结果，针对一个朴素的 PCI 风格传输预测：原始 PCI-A 在工作区代理中被反转（表 3）。我们认为这种反演是有益信息而非尴尬的：在工程瓶颈系统中，更紧密的耦合和组织可以降低基于可压缩性的轨迹复杂度，而非结构化失效动力学则可能使轨迹复杂度膨胀。虽然ΔPCI 提供了适度的对比“选择复杂性”信号，但它仍是一种启发式诊断（非Φ的直接检验），其行为依赖于故障/噪声对轨迹的影响。未来工作应实现更直接的Φ近似和/或更适合软件代理的整体与部分整合指标，并应明确测试哪些扰动和读出选择恢复了预期的 IIT 对齐单调性。

第三，我们的任务是相对简单的网格世界，具有离散动作和有限的状态空间。在更丰富的连续控制环境中，广播-放大/噪声衰减的区别可能以不同方式表现出来，噪声可能在更长时间范围内积累。

去噪归因控制（示波器）。我们包含仅 HOT 和仅 BC 的压缩控制（图 4），以加强实验 3 鲁棒性结果的归因。然而，这些控制是作为试点运行（n = 5 个种子），并且涵盖有限的压缩机家族（线性投影、小 MLP、固定随机投影）。部分压缩机基线在基线精度上表现不佳，限制了其鲁棒性阈值的可解释性。未来工作应扩展这些归因控制（更多种子;参数匹配变体;额外的瓶颈配置;以及更强的学习压缩器，如自编码器式重建目标），以测试结论在更广泛架构变异下是否稳定。

第四，我们的代理通过专家预言机的行为克隆训练，实验 1 包含一个明确的辅助损失来监督信心头。这引发了一个潜在担忧，即自模型可能仅仅学会预测神谕的行为，而非监控主体自身的认知状态。有两个澄清很重要。（i） oracle 仅用于生成训练目标;推断时从不参考该指标，所有报告的效应均通过训练后扰动（病灶、容量减少和潜在工作区噪声）下的训练代理人自主部署来衡量，这些扰动相对于示范分布不一致。（ii） 元认知监督目标由代理自身预期报告决策的正确性定义（代理日志的最大值与该步骤 Oracle 提供的基准标签的比较）。因此，自模型被训练为回答“鉴于我内部状态，我当前的决策是否正确？”而非预测预言机是否正确。尽管如此，强化学习可以以质的不同方式与相同架构互动（探索、学分分配和稳定性），而确定在强化学习下是否会出现类似的分离（特别是是否能仅通过奖励而无标记正确性而学习校准的元认知）仍是一个开放且重要的未来方向。

最后，我们的代理人在任何现象学意义上都没有意识;它们是理论的参考实现。B2 通过 HOT 测试却没有“感受”任何东西，这凸显了功能指标与主观体验之间的差距。然而，我们认为功能验证是理解意识的必要（即使不充分）步骤：如果某一机制即使在简化的合成系统中也未能产生预测的功能后果，那么它很可能是对生物现象的正确解释。

未来工作将探讨这些架构组件在更大时间冲突下如何相互作用，测试广播与元认知之间的层级关系是否推广到语言模型及其他现代人工智能架构，并探索在无明确监督下维持元认知稳定性的机制。最终，目标是确定广播、整合和元认知这三位一体是否能作为跨不同领域中普遍智能和适应力的稳健预测因子。

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斯坦尼斯拉斯·德哈恩、让-皮埃尔·尚格、莱昂内尔·纳卡什、杰罗姆·萨库尔和克莱尔·塞尔让。有意识，
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